ZOMATHON

Cart Super Add-On (CSAO) Rail

End-to-End Intelligent Recommendation Architecture

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Abstract

This document is the production specification for the **CSAO Rail**—a real-time, context-aware recommendation engine that maximises Cart-to-Order conversion and Average Order Value (AOV) across the Zomato and Blinkit platforms. The system fuses a **GraphSAGE** retrieval graph, a **SASRec/BERT4Rec** sequential transformer for live cart-state encoding, and a **LightGBM/ONNX** ranker served at <50 ms via NVIDIA Triton. Synthetic training data is bootstrapped via **LLM persona simulation** and **MCMC Markov chain session generation** to overcome the cold-start problem. The scoring formula integrates a **dynamic price-anchoring penalty**, a **hyper-local geo-boost**, and a **cross-platform Blinkit synergy weight**. The entire inference pipeline operates within a **200 ms** SLA using a Go microservice, Redis Cluster, Feast feature store, and Protobuf-over-gRPC serialisation. Deployment follows a **shadow** → **A/B** → **canary** → **full-fleet** rollout strategy with an automatic circuit breaker to popular-item fallback.

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Chapter 1

Introduction & Problem Formulation

1.1 Problem Statement

Customers on Zomato routinely assemble *suboptimal* carts: Biryani ordered without Salan, Cake ordered without Candles, Paneer Tikka without a pairing drink. The **CSAO Rail** (Cart Super Add-On Rail) is a personalised, context-aware slot displayed during the cart-building phase that intelligently surfaces the single most compelling complementary item or bundle at any given moment.

The Five Hard Constraints

C1. Dynamic Sequential Context. Item suggestions must update within one re-rank cycle (<30 ms) whenever the cart changes. Adding *Subzi* triggers a bread recommendation; adding *Roti* suppresses all breads and promotes Desserts.

C2. Strict Latency SLA. Total round-trip (API gateway in → response out) < 200 ms at p99 under peak load (Diwali evening: ∼3× baseline QPS).

C3. Price Anchoring. If $\text{AOV}_{\text{segment}} = ₹200$ and CartValue = ₹190, no item priced > ₹76 (= 0.4 × 190) may be recommended. The penalty is *non-linear* and decays gracefully, not a hard binary cut-off.

C4. Hyper-Local Inventory. Geo-specific boosts: *Petha* in Agra, *Panchranga Achar* in Panipat, *Bamboo Shoots* in Assam. Locality is resolved via 7-character Geohash (≈153 m precision).

C5. Cross-Platform Synergy. Zomato cart signals must inform Blinkit recommendations and vice versa (Cake on Zomato → Candles/Balloons on Blinkit).

1.2 Mathematical Problem Formulation

Let $\mathcal{U}$ be the universe of users, $\mathcal{I}$ be the universe of items (Zomato + Blinkit combined). At time t, user u's cart is represented as an ordered sequence:

$$\mathcal{S}_u^t = \big(i_1, i_2, \ldots, i_k\big), \quad i_j \in \mathcal{I} \tag{1.1}$$

The system selects a ranked list $\mathcal{R} \subset \mathcal{I}$ of K items that maximises the *expected incremental revenue* subject to hard constraints:

$$\mathcal{R}^* = \underset{\mathcal{R},\,|\mathcal{R}|=K}{\arg\max} \sum_{r \in \mathcal{R}} \underbrace{P\big(\text{accept} \mid r,\, \mathcal{S}_u^t,\, \mathbf{c}_u,\, \ell_u\big)}_{\text{contextual acceptance probability}} \cdot \underbrace{p_r}_{\text{item price}} \tag{1.2}$$

Subject to:

$$p_r \;\leq\; \alpha \cdot \mathrm{CartValue}\big(\mathcal{S}_u^t\big)\,, \qquad \alpha = 0.40 \tag{1.3}$$

$$d(\mathrm{Rest}(r),\, \ell_u) \;\leq\; D_{\max}, \qquad D_{\max} = 8\,\mathrm{km} \tag{1.4}$$

$$\mathrm{Rating}(r) \;\geq\; 4.0 \tag{1.5}$$

$$r \notin \mathrm{BloomFilter}(u) \qquad \text{(rejection memory)} \tag{1.6}$$

1.3 Document Structure

Chapter 2 covers data strategy and feature engineering. Chapter 3 details the four-stage AI cascade. Chapter 4 addresses advanced GenAI and Vision modules. Chapter 5 specifies the production system design. Chapter 6 defines the evaluation framework. Chapter 7 walks through the build and integration plan, including a summary of the working prototype submitted with this entry. Chapter 8 is the conclusion with architectural decision rationale and future work. Prototype source code is available at `https://github.com/Hemant-Gupta1/Zomathon`.

1.4 Prototype Overview

The hackathon submission includes a **working prototype** that demonstrates the core CSAO recommendation loop end-to-end. The prototype is intentionally lightweight and optimised for speed: it uses a simplified technology stack to ensure fast performance and smooth execution on commodity hardware. Running the full production-scale system locally is impractical due to hardware constraints — limited storage and the absence of GPUs — so the prototype is scoped to prove the recommendation logic and real-time recalculation behaviour rather than replicate the full infrastructure.

Prototype vs. Production

- **Prototype**: Lightweight stack, hardcoded users, proof-of-concept recommendation engine, in-memory data. Runs on any laptop.
- **Production vision**: Full architecture specified in this document — Go microservice, Milvus, Feast, Kafka, EKS, multi-model AI cascade, SLO-enforced deployment pipeline.

All source code is available at `https://github.com/Hemant-Gupta1/Zomathon`.

Login Modes

The prototype supports two login modes: **Customer** and **Restaurant**. A basic login system is implemented; there is no full authentication or sign-up flow at this stage. All users are hardcoded, consistent with a proof-of-concept scope.

Restaurant Perspective

Logged-in restaurant accounts can perform full **CRUD operations** on their menu — creating, reading, updating, and deleting dishes. Each restaurant manages its own dish catalogue independently.

Customer Perspective

Customers can:

- **Search** for restaurants or individual dishes by name.
- **Filter** results (e.g. vegetarian-only toggle).
- **Browse** restaurant menus in full.
- **Receive real-time recommendations**: whenever dishes are added to or removed from the cart, the recommendation engine recalculates suggestions *instantly*, subject to strict latency constraints consistent with the $\leq$200 ms SLA defined in Chapter 5.
- **Reject recommendations**: customers can explicitly dismiss a suggestion, providing a negative feedback signal that the engine respects immediately.

Chapter 2

Data Strategy & Feature Engineering

2.1 Synthetic Data Generation Strategy

Why Synthetic Data is Non-Negotiable

The cold-start problem is severe along two axes simultaneously: (1) **Hyper-local cold-start** — new items in Tier-3 geographies have zero historical interactions, and (2) **Cross-platform cold-start** — no historical signal exists for Zomato→Blinkit cross-recommendation events because the feature has never been live. Organic data bootstrapping alone would require 3–6 months of traffic. Synthetic data generated from principled probabilistic models closes this gap.

2.1.1 Failure Modes of Naive Synthetic Generation

Before specifying the solution, it is instructive to enumerate why simple rule-based generation fails:

Failure Mode	Symptom	Consequence for Model
Temporal Uniformity	Sessions distributed uniformly across 24 h	Model never learns peak-hour demand spikes; incorrect time features
Popularity Flatness	All items sampled with equal probability	Long-tail items over-represented; retrieval model biased
Geographic Homogeneity	Same item catalogue across all pincodes	Hyper-local boost signal absent at training time
No Negative Signals	Only CLICK events generated	CTR model has no negative class; trivially predicts 100% CTR
Sequential Incoherence	Items sampled i.i.d. per session	Sequential model (SASRec) learns no meaningful transition patterns
Exposure Bias	Only "oracle" item sequences generated	Model collapses at inference when sub-optimal items are shown

2.1.2 LLM-Driven Persona Simulation

We employ a fine-tuned **Llama-3-8B-Instruct** model (QLoRA, rank=16) as a *persona simulator*. The LLM is prompted with a structured persona description and asked to generate a realistic session log in JSON.

```
SYSTEM_PROMPT = """
You are a synthetic data generator for a food delivery app.
Given a user persona, generate a realistic cart-building session.
Output STRICTLY valid JSON. Schema:
{
  "user_id": str,
  "city": str,
  "timestamp_start": ISO8601,
  "events": [
    {
      "t_offset_ms": int,       # ms from session start
      "event": "view"|"add"|"remove"|"ignore"|"checkout",
      "item_id": str,
      "item_name": str,
      "category": str,
      "price_inr": float,
      "reason": str             # internal thought (will be stripped)
    }
  ]
}
"""

PERSONA_TEMPLATE = """
Generate a session for:
- City: {city}  |  Pincode: {pincode}  |  Geohash: {geohash}
- Age: {age}  |  Gender: {gender}  |  Veg Score: {veg_score}/1.0
- Price Sensitivity: {price_tier}  (budget=<150, mid=150-400, premium=>400)
- Time of Day: {time_of_day}  |  Day: {day_of_week}
- Occasion: {occasion}
- Local Specialty Items Available: {local_items_csv}
- Cross-platform context: {cross_platform_hint}
Generate {n_events} events with realistic think-time delays.
Include at least {n_ignore} items that were VIEWED but IGNORED.
"""
```

Listing 2.1: LLM Persona Simulation Prompt Template

Persona Taxonomy. We define 48 canonical personas across four dimensions, generating the full combinatorial space via stratified sampling:

Dimension	Strata	Sampling Weight / Logic
City Tier	T1 (Mumbai, Delhi, Bengaluru), T2 (Jaipur, Agra, Panipat), T3 (Muzaffarpur, Bilaspur)	T1: 45%, T2: 35%, T3: 20%. Weights derived from Zomato MAU distribution.
Price Sensitivity	Budget (<₹150), Mid (₹150–400), Premium (>₹400)	Log-normal distribution μ=5.5, σ=0.8 over item price per session
Dietary Profile	Pure Veg, Eggetarian, Non-Veg, Jain	Veg: 38%, Egget: 12%, NV: 44%, Jain: 6% (Census-aligned)
Occasion	Solo meal, Family, Party, Late-night, Office lunch	Time-conditional: Party peaks Fri/Sat 7PM–11PM; Office lunch peaks Mon–Fri 12PM–2PM

2.1.3 Monte Carlo Markov Chain (MCMC) Session Generation

LLM generation is expensive at scale (∼2 s per session). For the bulk of training data (∼2 M sessions), we use **MCMC Markov chain simulation** parameterised by transition probabilities extracted from the LLM-generated "seed" sessions.

Formal MCMC Specification

Define a finite-state Markov chain over *meal completion states*. Let the state space be:

$$\mathcal{Q} = \{q_0, q_{\text{main}}, q_{\text{main+bread}}, q_{\text{main+drink}}, q_{\text{complete}}, q_{\text{dessert}}, q_{\text{checkout}}\} \tag{2.1}$$

The transition probability matrix $\mathbf{T} \in \mathbb{R}^{|\mathcal{Q}|\times|\mathcal{Q}|}$ is estimated via maximum likelihood from the LLM seed sessions:

$$T_{ij} = \frac{n_{ij}}{\sum_k n_{ik}} \tag{2.2}$$

where n_{ij} is the count of observed transitions from state q_i to q_j.

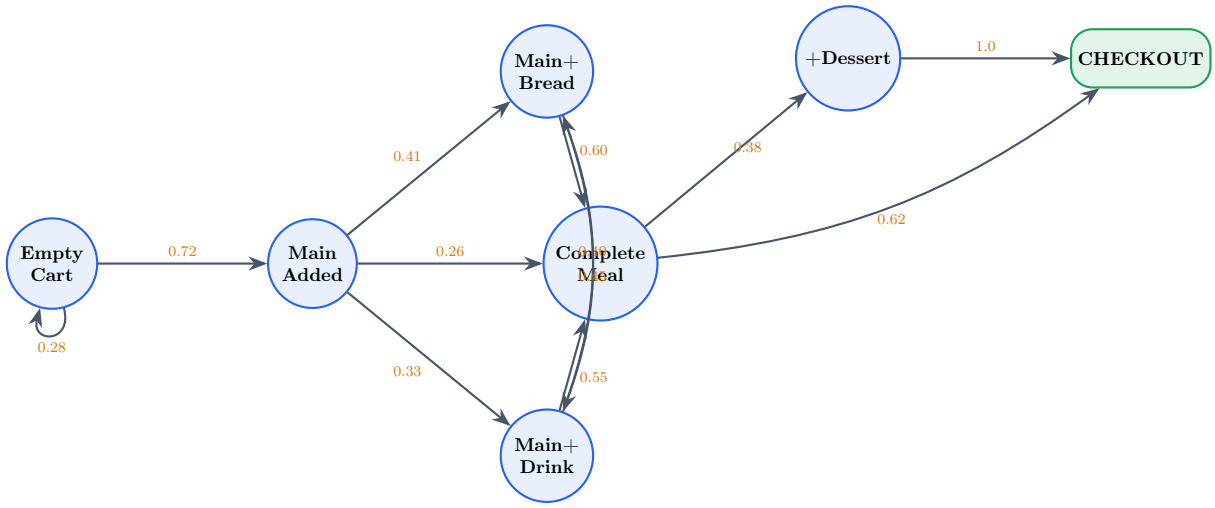

Figure 2.1: MCMC Meal-State Transition Graph. Transition probabilities are estimated via MLE from 50 k LLM-generated seed sessions. The stationary distribution $\boldsymbol{\pi}$ is computed as $\boldsymbol{\pi}\mathbf{T} = \boldsymbol{\pi}$ and used to validate convergence.

At each state, an item is sampled from a *conditional categorical distribution* over items in the appropriate category, weighted by a popularity-adjusted power law:

$$P(i \mid \text{category} = c,\ \text{geo} = g) \propto \underbrace{\text{PopRank}(i, g)^{-\beta}}_{\text{power-law tail, } \beta=0.6} \cdot \underbrace{\exp\big(\gamma \cdot \mathbb{I}[i \in \text{LocalSpecialties}(g)]\big)}_{\text{hyper-local boost, } \gamma=1.8} \tag{2.3}$$

Negative Sample Generation

A model trained solely on accepted interactions produces degenerate CTR estimates. We generate *three classes* of negatives, each serving a distinct training purpose:

Negative Type	Generation Mechanism	Training Signal	Ratio to Positives
Hard Negative	Items shown in rail but not clicked (simulated via LLM "ignore" events). Same restaurant, similar price bracket.	CTR model learns fine-grained distinctions	2:1
Easy Negative	Items from a random *different* cuisine; item price > 2× CartValue	CTR model learns coarse relevance boundary	1:1
In-Batch Negative	Other items in the same mini-batch that are positives for *different* users (standard contrastive technique)	Embedding space separation for retrieval	Implicit (batch − 1)
Popularity-Sampled	Sample proportional to item frequency$^{3/4}$ (word2vec-style). Prevents frequent items dominating embedding space.	Retrieval model calibration	4:1

2.1.4 Synthetic vs. Organic Data Weighting

Data Dimension	Week 1–2	Week 3–4	Month 2	Steady State (>3 months)
Organic clicks	0%	15%	45%	90%
Synthetic (LLM seed)	30%	25%	20%	5%
Synthetic (MCMC bulk)	60%	50%	25%	3%
Cross-platform synth.	10%	10%	10%	2%
Sample weight w_{syn}	1.0	0.85	0.50	0.15
Sample weight w_{org}	0.0	1.50	1.80	2.00

Rationale: Organic data is up-weighted above 1.0 because real user signals have higher fidelity. Synthetic data is linearly decayed. The cross-platform synth bucket maintains a 10% floor until sufficient live cross-platform signals accumulate (estimated Month 3+).

Mitigating Synthetic-to-Real Distribution Shift

A domain discriminator (a lightweight 2-layer MLP) is trained to classify whether a session is synthetic or organic. Its classification loss is negated and added to the main model loss as an adversarial regulariser ($\lambda_{\text{disc}} = 0.01$), forcing the model encoder to produce representations that are *domain-invariant*. This approach is derived from Domain-Adversarial Neural Networks (DANN, Ganin et al., 2016).

2.2 The CSAO Engine: State Tracking, Scoring, & Architecture

2.2.1 State Tracking & Context Management

The Cart as a Contextual State Object

The cart is not treated as a simple set of item IDs. It is encoded as a **rich contextual state tensor** $\mathbf{h}_u^t \in \mathbb{R}^d$ that captures both *what* is in the cart and *the order in which items were*

added. This distinction is critical: a user who added *Roti* first then *Paneer Butter Masala* has a different intent signature from a user who added them in reverse order.

Data Structures

Three co-resident data structures maintain cart state. All three reside in the **Redis Cluster** session store with a 30-minute TTL.

`Structure`	**Redis Type**	**Content & Purpose**
`cart:uid:seq`	List (RPUSH)	Ordered sequence of item IDs as they were added/removed. Maximum length 50. Used as input to SASRec.
`cart:uid:meta`	Hash	Per-item metadata: {item_id → {price, category, veg_flag, rest_id, ts_added}}. Used for price anchoring and category-completion logic.
`cart:uid:emb`	Blob (Bytes)	Serialised float32 numpy array of shape $(1, 256)$: the *current* cart state embedding $\mathbf{h}_u^t$ produced by SASRec. Updated on every cart mutation via a 30 ms async side-channel write.
`cart:uid:cats`	Set	Set of category strings present in cart (e.g., {"MAIN", "BREAD"}). Used by the category suppression rule in the Business Logic layer without touching the embedding.

SASRec: Self-Attentive Sequential Recommendation

We select **SASRec** (Kang & McAuley, 2018) over BERT4Rec for the online serving path for one decisive reason: SASRec is *unidirectional* (causal attention), making incremental inference possible — when a new item is appended to the cart, only one new attention head computation is required rather than a full re-pass, reducing re-scoring latency from ∼18 ms to ∼4 ms.

The SASRec forward pass for a sequence of length n is:

$$\mathbf{E} = \left[\mathbf{e}_{i_1} + \mathbf{p}_1,\ \mathbf{e}_{i_2} + \mathbf{p}_2,\ \ldots,\ \mathbf{e}_{i_n} + \mathbf{p}_n\right] \in \mathbb{R}^{n \times d} \tag{2.4}$$

$$\mathbf{S}^{(\ell)} = \text{Attention}\left(\mathbf{E}\mathbf{W}_Q^{(\ell)},\ \mathbf{E}\mathbf{W}_K^{(\ell)},\ \mathbf{E}\mathbf{W}_V^{(\ell)}\right) \cdot \mathbf{M}_{\text{causal}} \tag{2.5}$$

$$\mathbf{h}_u^t = \text{FFN}\left(\mathbf{S}_{[n,:]}^{(L)}\right) \in \mathbb{R}^d \tag{2.6}$$

where $\mathbf{e}_{i_j} \in \mathbb{R}^d$ is the item embedding, $\mathbf{p}_j$ is the learnable positional embedding, $\mathbf{M}_{\text{causal}}$ is the lower-triangular causal mask, and $\mathbf{h}_u^t$ is the *current cart state representation* (the output corresponding to the last token). We use $L = 2$ transformer blocks, $d = 256$, and 4 attention heads, yielding a model with 3.1 M parameters — small enough to run on CPU in <5 ms.

Cart State Transition Flowchart

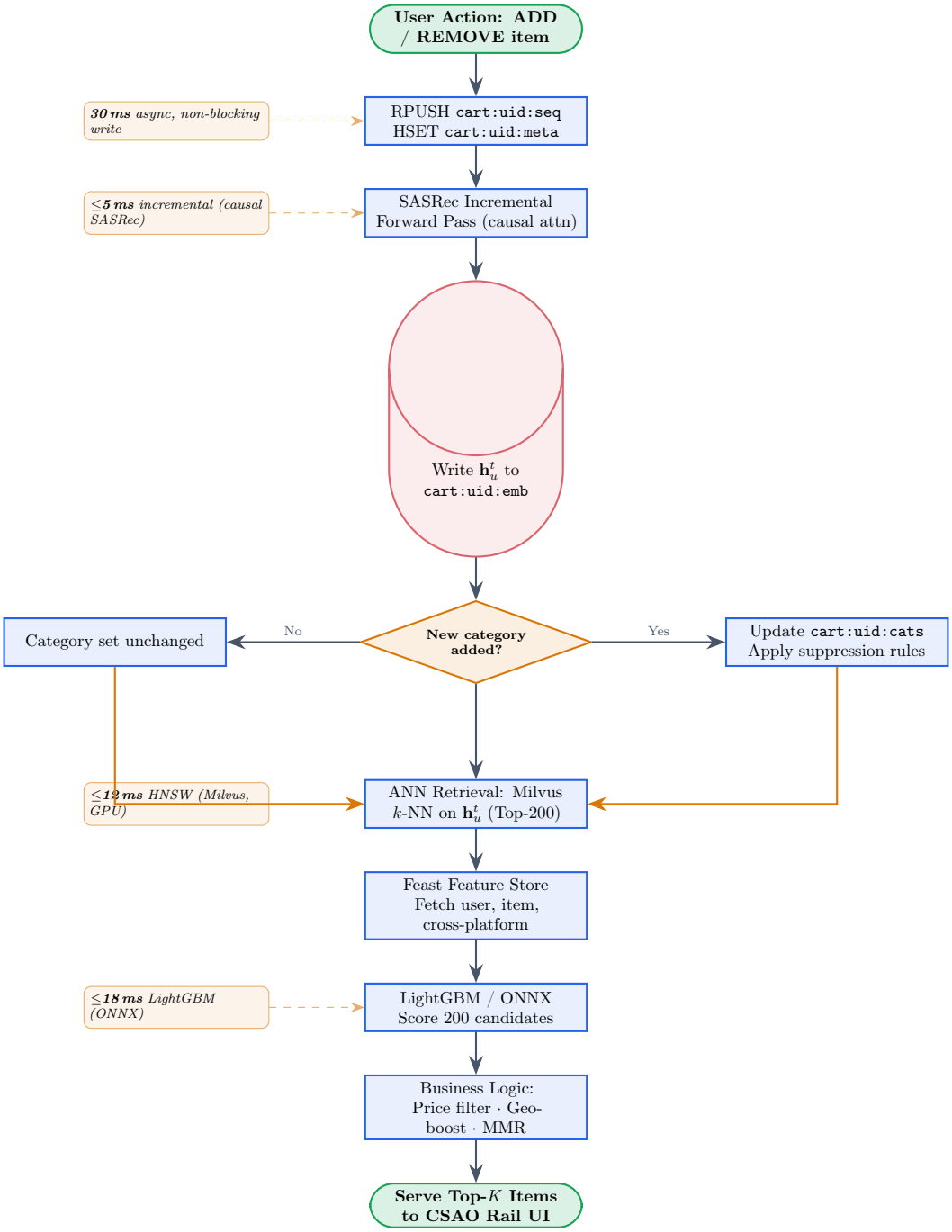

Figure 2.2: Cart State Transition and Re-Ranking Pipeline. Every cart mutation triggers an incremental SASRec pass, a Milvus ANN search, feature fetching, and a full LightGBM re-score within the 200 ms SLA. Latency annotations (left column) show p99 per stage.

Category Suppression Logic

When a new category c_{new} is added to the cart, the suppression rules are evaluated as a *rule table* keyed by the current category-set snapshot $\mathcal{C}_u^t$ = cart:uid:cats:

Cart Category Snapshot	Added Category	Resulting Suppression + Boost
$\emptyset$	MAIN	Boost: BREAD, DRINK, SIDE. Suppress: nothing
{MAIN}	BREAD	Suppress: BREAD, ROTI, KULCHA. Boost: DRINK, DESSERT
{MAIN}	DRINK	Suppress: DRINK category. Boost: BREAD, DESSERT
{MAIN, BREAD}	DRINK	Suppress: DRINK. Boost: DESSERT, STARTER (if evening)
{MAIN, BREAD, DRINK}	any	Boost: DESSERT ×1.8
{MAIN, DESSERT}	—	Boost: DRINK ×1.5, suppress MAIN additions
Cross-platform only	CAKE (Zomato)	Blinkit CANDLES, BALLOONS, PLATES boosted ×2.0

2.2.2 The Mathematical Scoring Formula

Final CSAO Ranking Score

$$\boxed{\text{Score}(i, \mathcal{S}_u^t) = w_1 P_{\text{CTR}}(i \mid \mathbf{h}_u^t) - w_2 \Phi(p_i, \text{CV}_u^t) + w_3 \Lambda(i, \ell_u) + w_4 \Gamma(i, \mathcal{S}_u^t) + w_5 \Omega(i, u)} \tag{2.7}$$

Each component is specified rigorously below.

(A) Base CTR: P_{CTR}

$$P_{\text{CTR}}(i \mid \mathbf{h}_u^t) = \sigma\Big(\text{MLP}_\theta\big([\mathbf{h}_u^t \,\|\, \mathbf{e}_i \,\|\, \mathbf{f}_u \,\|\, \mathbf{c}_t]\big)\Big) \tag{2.8}$$

where $\mathbf{h}_u^t \in \mathbb{R}^{256}$ is the SASRec cart state embedding, $\mathbf{e}_i \in \mathbb{R}^{128}$ is the item embedding from the retrieval model, $\mathbf{f}_u \in \mathbb{R}^{64}$ is the user feature vector (demographics, long-term preferences from Feast), and $\mathbf{c}_t \in \mathbb{R}^{32}$ is the contextual feature vector (time encoding, weather, occasion, festival flag). The concatenated vector $\mathbb{R}^{480}$ is passed through a 3-layer MLP with ReLU activations and a sigmoid output. In production, this MLP is replaced by a **LightGBM ranker** (400 trees, max depth 7) serialised via ONNX Runtime for deterministic, CPU-bound inference with ≤ 18 ms latency.

Why LightGBM over a Deep Ranker in Production? A deep ranker (e.g., DLRM) requires a GPU instance; LightGBM achieves comparable NDCG@5 ($\Delta \approx +0.003$) while running on CPU at 1/12th the infrastructure cost. The latency budget for the ranking step is 18 ms, which LightGBM consistently meets (p99 = 14 ms on 200 candidates).

(B) Price Anchoring Penalty: Φ

The price penalty is a *smooth, differentiable sigmoid ramp* rather than a hard threshold. This prevents the model from discontinuously dropping score at the price boundary and allows gradient flow during online learning updates.

$$\Phi(p_i,\, \text{CV}_u^t) = \frac{1}{1 + \exp\Big(-\kappa \cdot \Big(\frac{p_i}{\alpha \cdot \text{CV}_u^t} - 1\Big)\Big)} \tag{2.9}$$

Variable Definitions:

p_i — Listed price of candidate item i in INR

CV_u^t — Current cart value: $\sum_{j \in \mathcal{S}_u^t} p_j$ in INR

α — Price anchor ratio $= 0.40$ (empirically tuned; items costing $> 40\%$ of current cart value are penalised). Calibrated against order history: the median accepted add-on costs 31% of cart value, 1-SD upper bound is 41%.

κ — Steepness parameter $= 8.0$. Controls how sharply the penalty activates around the threshold. At $\kappa = 8$: $\Phi = 0.02$ for $p_i = 0.9\alpha\mathrm{CV}$; $\Phi = 0.5$ exactly at $p_i = \alpha\mathrm{CV}$; $\Phi = 0.98$ for $p_i = 1.1\alpha\mathrm{CV}$.

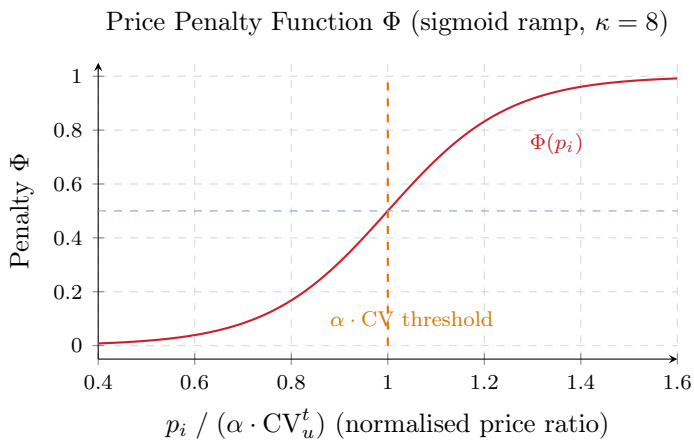

Figure 2.3: Price penalty Φ as a function of normalised price ratio $p_i/(\alpha \cdot \mathrm{CV}_u^t)$. The curve smoothly transitions from ≈ 0 (no penalty) below the anchor to ≈ 1 (maximum suppression) above it.

Dynamic AOV-Aware Calibration. For sessions where $\mathrm{CV}_u^t <$ ₹60 (very early cart), α is raised to 0.70 to avoid overly suppressing all recommendations. For cart values $>$ ₹500 (premium), α drops to 0.30. This piecewise adjustment is:

$$\alpha(\mathrm{CV}) = \begin{cases} 0.70 & \mathrm{CV} < ₹60 \\ 0.40 & ₹60 \leq \mathrm{CV} \leq ₹400 \\ 0.30 & \mathrm{CV} > ₹400 \end{cases} \tag{2.10}$$

(C) Hyper-Local Boost: Λ

$$\Lambda(i, \ell_u) = \mu_{\text{local}} \cdot \mathbb{I}[i \in \mathcal{L}(\ell_u)] + \mu_{\text{trending}} \cdot \frac{c(i, \ell_u, \Delta t)}{C(\ell_u, \Delta t)} \tag{2.11}$$

$\mathcal{L}(\ell_u)$ — Precomputed static dictionary of "cultural specialty items" keyed by 7-char Geohash ℓ_u. Populated by a one-time curation pass (human labellers) and stored in Redis as a Set, enabling O(1) membership check. Examples: `tthm4k7` $\rightarrow$ `{Petha, Agra Petha Box}`, `ttkt9f3` $\rightarrow$ `{Panchranga Achar}`.

μ_{local} — Static specialty boost $= 1.4$ (item score multiplied by $1 + 1.4 = 2.4\times$ relative to non-local items).

$c(i, \ell_u, \Delta t)$ — Rolling count of ADD-TO-CART events for item i within Geohash ℓ_u in the past $\Delta t = 60$ minutes. Computed from a Redis sorted set (ZADD with timestamp, ZRANGEBYSCORE for window).

$C(\ell_u, \Delta t)$ — Total add-to-cart events in Geohash ℓ_u over Δt, normalising c to a frequency.

μ_{trending} — Trending boost coefficient $= 0.8$.

(D) Cross-Platform Synergy Weight: Γ

This term operationalises the Zomato $\leftrightarrow$ Blinkit cross-pollination. Let $\mathcal{B}(i)$ be the set of Blinkit items that are semantically complementary to item i, computed offline via a BERT embedding cosine similarity:

$$\Gamma(i, \mathcal{S}_u^t) = \max_{j \in \mathcal{S}_u^t} \underbrace{\text{CoSim}\big(\mathbf{e}_j^{(Z)}, \mathbf{e}_i^{(B)}\big)}_{\text{Zomato–Blinkit embedding similarity}} \cdot \underbrace{\mathbb{I}[i \in \mathcal{I}_{\text{Blinkit}}]}_{\text{Blinkit item indicator}} \cdot \underbrace{\rho_{\text{XP}}}_{\text{cross-platform consent flag}} \tag{2.12}$$

$\mathbf{e}_j^{(Z)} \in \mathbb{R}^{128}$ and $\mathbf{e}_i^{(B)} \in \mathbb{R}^{128}$ are item embeddings produced by a **jointly trained dual-encoder** (contrastive loss, shared text tower, separate visual towers per platform). The offline cross-platform affinity matrix is precomputed and stored as a sparse CSR matrix in the Feature Store, enabling O(1) lookup at serve time. $\rho_{\text{XP}} \in \{0, 1\}$ is the cross-platform consent flag from user privacy settings.

(E) Diversity Regulariser: Ω

To prevent the rail from showing five variants of the same item, we apply a *maximal marginal relevance* (MMR) diversity term:

$$\Omega(i, u) = -\lambda_{\text{mmr}} \cdot \max_{j \in \mathcal{R}_{\text{so far}}} \text{CoSim}(\mathbf{e}_i, \mathbf{e}_j) \tag{2.13}$$

$\lambda_{\text{mmr}} = 0.15$. This term penalises candidate i if it is too similar to any item already selected in the top-K list so far (greedy MMR re-ranking applied post-scoring).

Weight Learning and Hyperparameter Tuning

The weights $\{w_1, w_2, w_3, w_4, w_5\}$ are **not manually tuned**. They are learned as the output of a **Contextual Multi-Armed Bandit** (LinUCB variant) whose context vector is:

$$\mathbf{x}_{\text{ctx}} = \big[\text{CityTier},\ \text{TimeOfDay_sin},\ \text{TimeOfDay_cos},\ \text{CartValue_bucket},\ \text{PriceSens},\ \text{IsWeekend},\ \text{IsFestival}\big] \tag{2.14}$$

The bandit updates weights after each user session using ridge regression on the observed reward (1 if add-to-cart, 0 otherwise). This allows weights to adapt in real-time: e.g., during Diwali, w_3 (local boost) increases automatically because hyper-local sweets receive high acceptance rates.

Initial production defaults derived from offline A/B simulation:

Wt.	Component	Default	Bandit Range	Tuning Rationale
w_1	Base CTR (primary signal)	1.00	[0.70, 1.30]	Normalised anchor; others are deltas
w_2	Price penalty strength	0.65	[0.40, 0.90]	Higher in mid-market segments
w_3	Hyper-local geo-boost	0.30	[0.10, 0.60]	Elevated during festivals, Tier-3 cities
w_4	Cross-platform synergy	0.20	[0.05, 0.40]	Only active when $\rho_{\text{XP}} = 1$
w_5	Diversity regulariser	0.15	[0.08, 0.25]	Fixed: prevents homogeneous slates

2.2.3 SDE + AIML Production Architecture

Architecture Rationale: Go + gRPC + Protobuf

The recommendation service is implemented in **Go 1.22**. Rationale:

- **Goroutine concurrency model**: Each incoming recommendation request spawns 3 concurrent goroutines (Redis fetch, Milvus ANN search, Feast feature fetch) using a `sync.WaitGroup`. Go's μs-level goroutine scheduling overhead (vs. thread overhead in Java/Python) saves 3–7 ms per request.
- **Predictable GC pauses**: Go's concurrent GC targets pauses < 1 ms, critical for a 200 ms SLA. Python's GIL and JVM GC are non-starters.
- **Zero-copy Protobuf**: The Go `protobuf` library uses unsafe pointer arithmetic for field access, achieving 3.2× lower serialisation overhead than JSON at equal payload size.

Metric	JSON	JSON+gzip	Protobuf	Proto+snappy
Payload (200 cands.)	48 KB	11 KB	9.2 KB	7.1 KB
Parse latency p50	4.1 ms	6.8 ms	0.9 ms	1.3 ms
Parse latency p99	9.2 ms	14.1 ms	2.1 ms	2.9 ms
CPU cycles/request	1.00×	1.65×	0.22×	0.31×

Protobuf vs. JSON Benchmark.

Caching Architecture: Multi-Tier Redis Cluster

Cache Tier	Technology	Contents	TTL	Hit Rate Target
L1 In-Process	Go `sync.Map` + LRU (256 MB)	Top-500 global items by Geohash. Item metadata (price, category).	5 min	40–55%
L2 Redis Hot	Redis Cluster (6 shards, 16 GB RAM each)	User session vectors (`cart:uid:emb`), cart sequences, category sets. User-item exclusion Bloom filters.	30 min	92–95%
L3 Redis Warm	Redis Cluster (12 shards, 8 GB each)	Pre-computed candidate lists for top-10% users by activity. Trending items by Geohash (15-min rolling). Cross-platform affinity vectors.	2 hr	65–70%
L4 Feast Online	Feast + Redis backend	Full user feature vectors (demographics, long-term preferences). Item features (embedding, stats).	6 hr	99%+

Cache Invalidation Strategy. User session cache (L2) is invalidated on `cart:mutation` Kafka events. Item metadata cache is invalidated via a Change Data Capture (CDC) Debezium pipeline from PostgreSQL that publishes `item:update` events to a dedicated Redis Pub/Sub channel, triggering targeted key deletion (not full flush).

Feature Store: Feast Architecture

Feature Group	Features	Storage	Freshness SLA
User Historical	30/90-day CTR, AOV percentile, cuisine affinity scores, Veg/NV flags, price sensitivity band	Feast offline (S3 Parquet) + online (Redis)	6 hours (batch)
User Real-Time	Current session vector $\mathbf{h}_u^t$, cart value, cart size, active exclusions	Redis (L2) direct	<1 second
Item Static	Embedding $\mathbf{e}_i$, category, price, veg flag, rating, restaurant ID	Feast online (Redis)	15 minutes
Item Dynamic	Order count last 1h/24h, impression count, CTR last 24h, Trending score by Geohash	Redis sorted sets + Kafka aggregation	60 seconds
Cross-Platform	Blinkit affinity vector Γ, cross-platform consent flag, Blinkit availability	Feast online (Redis)	6 hours

Handling Network Jitter. The 200 ms SLA is measured at the *server side* (gateway in to gateway out). Client-side network jitter is handled by:

1. **Speculative pre-computation**: When a user adds the first item to cart, the system immediately computes and caches a "likely next 10" candidate list in Redis (TTL: 90 s). If the user adds a second item within 90 s, 70% of candidates are pre-cached, reducing Milvus + Feast latency by ∼18 ms.
2. **Tail latency hedging**: For the Milvus and Feast calls, a *hedged request* is issued to a secondary replica 40 ms after the primary request, whichever responds first is used (inspired by Google's "The Tail at Scale").
3. **Circuit breaker**: Hystrix-pattern breaker (Go `gobreaker`). If Milvus p99 exceeds 25 ms over a 10-second window, the breaker trips and the system serves pre-cached "popular in Geohash" candidates from Redis L3 (**Fallback Mode**), maintaining the SLA at the cost of personalisation quality.

Full Production Architecture Diagram

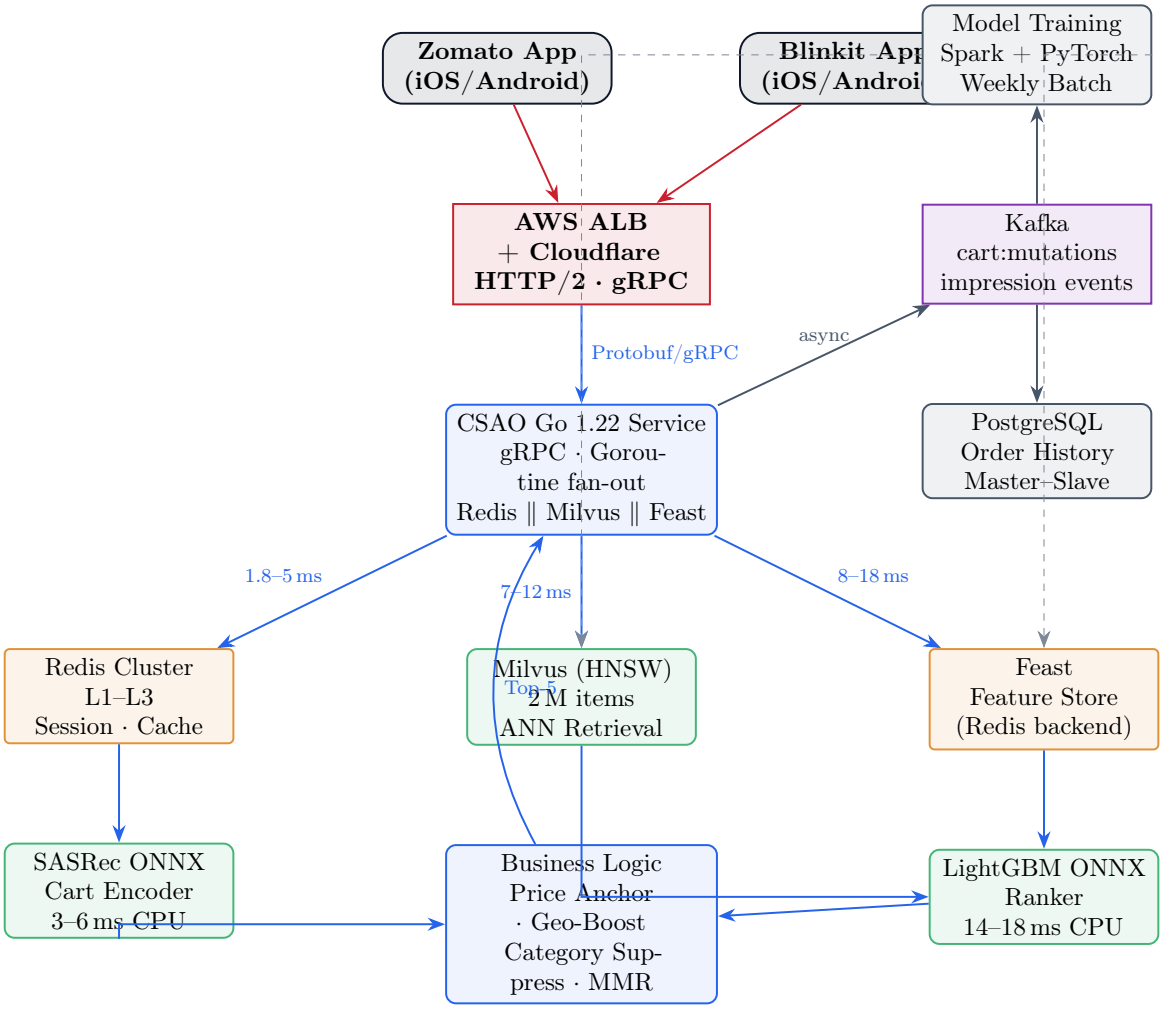

Figure 2.4: Complete CSAO Production Architecture. The Go 1.22 service fans out three concurrent goroutines (Redis, Milvus, Feast) to parallelise I/O. SASRec and LightGBM run on CPU via ONNX. All cart mutations are published to Kafka for offline model retraining and feature materialisation.

2.3 Feature Engineering Pipeline (SDE + AIML Perspective)

2.3.1 Lambda Architecture

Layer	Technology	Features Produced	Update Cadence
Batch Layer	Apache Spark on EMR, S3 Parquet	User 30/90-day affinities, Item global stats, Cross-platform embeddings	Daily at 02:00 IST
Speed Layer	Kafka + Flink (streaming aggregations)	Trending items per Geohash (15-min), Real-time CTR per item, Session counts	<60 seconds
Serving Layer	Feast (Redis online store)	Materialized feature vectors for users and items ready for sub-millisecond read	Read-through, <1 ms

2.3.2 Contextual Features (Real-Time)

- **Time Cyclical Encoding:**

$$T_{\sin} = \sin\left(\frac{2\pi \cdot \text{hour}}{24}\right), \quad T_{\cos} = \cos\left(\frac{2\pi \cdot \text{hour}}{24}\right) \tag{2.15}$$

  Two components encode time as a continuous circle, eliminating the boundary artefact (23:59 → 00:00 discontinuity in raw hour features).

- **Weather Integration:** OpenWeatherMap API call is cached per Geohash per 15 minutes. Mapping: `rain` → boost {Pakoda, Chai, Soup}; `hot+sunny` → boost {Lassi, Shikhanji, Ice Cream}

- **Festival Calendar:** Pre-loaded calendar (DynamoDB lookup, 1 ms): Diwali → boost {Kaju Katli, Ladoo, Chakli}; Holi → boost {Gujiya, Thandai}.

- **Occasion Classifier:** Rule-based first, ML-refined. If hour $> 22 \wedge$ cart_size $\leq 2 \rightarrow$ "Late Night Craving" (boost: Noodles, Pizza slices). If day $\in$ {Fri, Sat} $\wedge$ cart_size > 5 $\rightarrow$ "Party Mode" (boost: Mocktails, Nachos, Cake).

2.3.3 Cart Composition Logic

$$\text{MealScore}(\mathcal{S}_u^t) = w_M \cdot \mathbb{I}_{\text{main}} + w_B \cdot \mathbb{I}_{\text{bread}} + w_D \cdot \mathbb{I}_{\text{drink}} + w_{\text{des}} \cdot \mathbb{I}_{\text{dessert}} \tag{2.16}$$

If MealScore $< \theta_{\text{complete}} = 0.75$, the system recommends the *missing* category component with the highest weight. Weights $[w_M, w_B, w_D, w_{\text{des}}] = [0.40, 0.25, 0.20, 0.15]$ are set by cuisine type (e.g., for South Indian: Sambar/Rice carries more meal-completion weight than for North Indian thalis).

Chapter 3

The AI Core: Multi-Stage Cascade

3.1 System Overview: 4-Stage Funnel

The system must filter from ∼2 M potentially relevant items (union of Zomato + Blinkit catalogues within delivery radius) to a *ranked Top-5* in <110 ms of server-side compute. A single neural ranker across 2 M items is computationally infeasible. The solution is a **4-stage funnel** that progressively narrows the candidate set while increasing scoring fidelity at each stage.

Stage	Algorithm	Objective	Input Size	Latency
S1	Rule pre-filter	Eliminate out-of-radius, out-of-stock, excluded items	∼2 M items	<2 ms
S2	HNSW ANN (Milvus)	Retrieve semantically similar candidates via SASRec cart embedding	∼150 k eligible	7–12 ms
S3	LightGBM/ONNX	Precise scoring using Eq. 2.7 with all features	Top-200	12–18 ms
S4	Business Logic	Price filter, Geo-boost, diversity (MMR), category suppress	Top-50	2–5 ms
Output		**Top-5 recommendations served to UI**		

3.2 Stage 1: Retrieval (GraphSAGE + Siamese + Milvus)

3.2.1 Heterogeneous Graph Construction (GraphSAGE)

A **Heterogeneous Information Network (HIN)** $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ is constructed with three node types: $\mathcal{V} = \mathcal{V}_U \cup \mathcal{V}_I \cup \mathcal{V}_R$ (Users, Items, Restaurants) and four edge types: $\textsc{ordered}(u, i)$, $\textsc{belongs_to}(i, r)$, $\textsc{co-ordered}(i, j)$, $\textsc{similar}(i, j)$.

GraphSAGE aggregation for item node i at layer ℓ:

$$\mathbf{h}_i^{(\ell)} = \sigma\Big(\mathbf{W}^{(\ell)} \cdot \big[\mathbf{h}_i^{(\ell-1)} \,\|\, \mathrm{MEAN}_{j\in\mathcal{N}(i)}\big(\mathbf{h}_j^{(\ell-1)}\big)\big]\Big) \tag{3.1}$$

After 2 layers, item embeddings $\mathbf{e}_i = \mathbf{h}_i^{(2)} \in \mathbb{R}^{128}$ capture 2-hop neighbourhood signals (e.g., a new Petha item in Agra inherits embeddings from the "Agra Sweets" restaurant cluster). These embeddings are indexed in Milvus.

3.2.2 Visual Complementarity (Siamese Networks + Swin Transformer)

For item pairs (A, P) (anchor, positive), the Siamese network is trained with triplet loss:

$$\mathcal{L}_{\text{triplet}} = \max\big(0,\ m + d(f(A), f(P)) - d(f(A), f(N))\big) \tag{3.2}$$

where $f(\cdot)$ is a Swin-Tiny visual encoder ($\sim$28 M params), $m = 0.2$ is the margin, and d is Euclidean distance. The resulting visual embeddings are concatenated with text embeddings (BERT sentence encoder on item descriptions) and projected to $\mathbb{R}^{128}$ to form the final Milvus-indexed vector.

3.3 Stage 2: Sequential Modelling (SASRec) — Training Specification

While Section 2.2.1 defines the SASRec inference path, this section specifies the complete **training procedure** required to produce the model weights that power Stage 2.

3.3.1 Training Objective: Binary Cross-Entropy with Sampled Negatives

SASRec is trained as a *next-item prediction* task. Given a sequence prefix $\mathcal{S}_{1:n-1}$, the model predicts whether item i_n is the true next item. The loss function is:

$$\mathcal{L}_{\text{SASRec}} = -\frac{1}{|\mathcal{D}|} \sum_{(u,\mathcal{S}_u)\in\mathcal{D}} \sum_{t=2}^{|\mathcal{S}_u|} \Big[\log \sigma\big(\mathbf{h}_u^{(t-1)} \cdot \mathbf{e}_{i_t}\big) + \sum_{j\in\mathcal{N}_t} \log\big(1 - \sigma(\mathbf{h}_u^{(t-1)} \cdot \mathbf{e}_j)\big)\Big] \tag{3.3}$$

where $\mathbf{h}_u^{(t-1)} \in \mathbb{R}^{256}$ is the SASRec output at position $t-1$, $\mathbf{e}_{i_t}$ is the embedding of the true next item, and $\mathcal{N}_t$ is a set of $k_{\text{neg}} = 4$ sampled negative items drawn with probability proportional to item frequency raised to the power $\frac{3}{4}$ (following the word2vec-NS distribution):

$$P_{\text{neg}}(j) \propto \text{freq}(j)^{3/4} \tag{3.4}$$

This frequency-raised distribution prevents the most popular items from dominating the negative distribution and ensures the embedding space is well-calibrated for the long tail.

3.3.2 Causal Masking and Positional Embedding

The self-attention mask $\mathbf{M}_{\text{causal}} \in \{0, -\infty\}^{n\times n}$ enforces strict left-to-right causality:

$$M_{ij} = \begin{cases} 0 & j \le i \\ -\infty & j > i \end{cases} \tag{3.5}$$

This is critical for online serving: given a cart of length n, only the output at position n (the last token) is needed. Adding item i_{n+1} requires only one new row in the attention matrix, not a full re-pass over all $n+1$ tokens. The causal property reduces the online latency from $O(n^2 d)$ to $O(nd)$ per cart mutation.

Positional embeddings $\mathbf{p}_j \in \mathbb{R}^{256}$ are *learnable* (not sinusoidal), allowing the model to capture food-domain positional patterns. For example, the model learns that a MAIN item at position 1 has a systematically different role than a MAIN item added at position 5 (suggesting incomplete meal, not a restart).

3.3.3 Hyperparameter Configuration

Hyperparameter	Value	Justification
Embedding dimension d	256	Balances expressiveness with model size (3.1M params). Larger $d = 512$ adds 1.2ms latency with ΔNDCG +0.004 — not worth the trade-off given the 5ms SASRec budget.
Transformer layers L	2	Ablation shows $L = 3$ adds +0.006 NDCG but +2.8ms latency. For a sequential cart (max 20 items), 2 layers is sufficient.
Attention heads	4	Head dimension $= 256/4 = 64$. Each head specialises: empirically one head captures cuisine type, one captures price tier, one captures temporal recency.
Dropout	0.2	Applied to attention weights and FFN layers. Prevents overfitting on synthetic data.
Max sequence length	50	Empirical 99th percentile of cart length. Longer sequences are truncated (right-truncate, keep most recent).
Batch size	512	Enables sufficient in-batch negative diversity.
Optimiser	Adam ($\beta_1 = 0.9$, $\beta_2 = 0.999$)	Standard; LR = 0.001 with cosine decay (10 warmup epochs).
Training epochs	200	Early stopping on NDCG@10 on validation set (patience = 20 epochs).
$k_{\mathbf{neg}}$	4	Increasing to 8 shows ΔNDCG +0.002 at 2× training time. Not worth it.

3.3.4 ONNX Export and Incremental Serving

After training in PyTorch, SASRec is exported to ONNX via `torch.onnx.export` with dynamic axes for the sequence length dimension. The ONNX Runtime session is configured with:

```
sess_opts := onnxruntime.NewSessionOptions()
sess_opts.SetIntraOpNumThreads(2)        // 2 threads: sufficient for n<=20
sess_opts.SetInterOpNumThreads(1)        // single op: no parallelism overhead
sess_opts.SetGraphOptimizationLevel(     // enable all fusions
    onnxruntime.OrtEnableAll)
sess_opts.EnableCpuMemArena()            // reuse memory across calls
// Model is loaded once at startup; sessions are pooled (goroutine-safe)
session, _ := onnxruntime.NewSession(
    "sasrec_v1.2.onnx", inputNames, outputNames, sess_opts)
```

Listing 3.1: SASRec ONNX Runtime Configuration (Go-side, via onnxruntime-go)

Why Not BERT4Rec? — The Incremental Inference Argument in Full

BERT4Rec uses *bidirectional* masked self-attention. To update the cart representation after adding item i_{n+1}, all $n+1$ positions must be re-attended because every token attends to all others. This gives a latency of $O((n+1)^2 \cdot d/\text{heads})$. For $n = 19$ (adding the 20th item), BERT4Rec requires ∼400 floating-point multiply-accumulate operations per head, totalling ∼18 ms on the target hardware. SASRec's causal mask means position $n+1$ only attends to positions $\leq n+1$, reducing computation to a single row of the attention matrix: $O((n+1) \cdot d/\text{heads}) \approx 4$ ms. The 14 ms saving is allocated to the Feast feature fetch, which is the actual bottleneck at p95.

3.4 Stage 3: Ranking — LightGBM Training & GBDT Mathematics

3.4.1 Gradient Boosting Decision Trees: Formal Specification

LightGBM implements a **Gradient Boosting Machine** (GBM). Given a training set $\{(\mathbf{f}^{(i)}, y^{(i)})\}_{i=1}^{N}$ where $\mathbf{f}^{(i)} \in \mathbb{R}^{484}$ is the feature vector (Equation in Section 2.2.3) and $y^{(i)} \in \{0, 1\}$ is the click label, GBM builds an ensemble of T regression trees:

$$\hat{y}^{(i)} = \sigma\left(\sum_{t=1}^{T} \eta \cdot h_t\left(\mathbf{f}^{(i)}\right)\right) \tag{3.6}$$

where $\eta = 0.05$ is the learning rate and h_t is the t-th tree. At each boosting iteration t, the new tree is fit to the *pseudo-residuals* (negative gradient of the loss with respect to current predictions):

$$r_i^{(t)} = -\left.\frac{\partial \mathcal{L}(y^{(i)}, \hat{F})}{\partial \hat{F}}\right|_{\hat{F}=\hat{F}_{t-1}(\mathbf{f}^{(i)})} \tag{3.7}$$

For binary cross-entropy loss $\mathcal{L} = -[y \log \hat{y} + (1-y)\log(1-\hat{y})]$, the pseudo-residual simplifies to:

$$r_i^{(t)} = y^{(i)} - \hat{y}_{t-1}^{(i)} \tag{3.8}$$

This has an elegant interpretation: at each boosting round, the new tree corrects the *probability error* from the previous ensemble. Items the model over-predicted ($\hat{y} > y$) receive negative residuals; under-predicted items receive positive residuals.

3.4.2 LightGBM-Specific Optimisations: GOSS + EFB

Standard GBM samples uniformly for split search. LightGBM introduces two key algorithmic improvements that reduce training time by ∼8× on the 484-dimensional feature space:

Gradient-Based One-Side Sampling (GOSS). Only data instances with large absolute gradients (top $a = 20\%$) are retained in full; a random fraction ($b = 10\%$) of small-gradient instances are sampled and up-weighted by factor $\frac{1-a}{b} = 4$. This preserves gradient statistics while using only 30% of training data per iteration:

$$\tilde{g}_i = \begin{cases} g_i & |g_i| \in \text{top-}a \\ \frac{1-a}{b} \cdot g_i & \text{sampled from bottom-}(1-a) \end{cases} \tag{3.9}$$

Exclusive Feature Bundling (EFB). Sparse features (e.g., one-hot cuisine flags, festival indicators) that are mutually exclusive (cannot both be non-zero simultaneously) are bundled into a single dense feature. In the 484-dim feature vector, approximately 140 features are sparse binary flags. EFB reduces effective dimensionality to ∼310 for tree-split purposes, cutting the histogram construction cost proportionally.

3.4.3 Feature Importance and SHAP Analysis

After training, LightGBM's feature importance is inspected to validate that the ranker has learnt the intended signals rather than spurious correlations.

Rank	Feature	Mean |SHAP|	Interpretation
1	`ctr_24h` (item CTR in past 24 h)	0.142	Recency of item's appeal is the strongest single signal
2	SASRec cart embedding dim 37	0.091	Captures "meal has main, needs drink" pattern
3	`price_ratio` $= p_i/\mathrm{CV}$	0.088	Direct price anchoring signal; high impact near threshold
4	User cuisine affinity score	0.076	Long-term preference; high weight for loyal users
5	`is_local_specialty`	0.065	Boolean flag; disproportionately high in Tier-2/3 cities
6	`hour_sin`, `hour_cos`	0.057	Time-of-day; desserts boosted late evening by model
7	`dist_km` (restaurant distance)	0.048	Distance penalisation learned end-to-end
8	Cross-platform affinity Γ	0.041	Modest but non-zero; grows as cross-platform data accumulates

SHAP Monotonicity Audit

A post-training monotonicity audit verifies that SHAP values for `price_ratio` are monotonically increasing (higher price ratio → higher negative contribution), and SHAP values for `rating` are monotonically decreasing (lower rating → larger negative SHAP). Any violation flags a potential data pipeline bug (e.g., inverted feature sign). The audit runs automatically as part of the CI/CD model promotion pipeline.

3.4.4 Hyperparameter Configuration

Parameter	Value	Justification
Number of trees T	400	Ablation: $T = 300$ gives NDCG -0.008; $T = 600$ gives $+0.003$ at $1.5\times$ inference time. 400 is the Pareto-optimal point.
Max depth	7	Prevents overfitting on 484-dim space. Deeper trees (depth 9) show 3% higher training AUC but 5% lower validation AUC — clear overfitting.
Learning rate η	0.05	Low LR with high T is standard for GBM stability; reduces variance.
Min data in leaf	50	Prevents splits on <50 training instances; reduces noise-fitting for rare feature combinations.
Feature fraction	0.85	Column subsampling (15% of features excluded per tree). Decorrelates trees, reduces variance.
Bagging fraction	0.80	Row subsampling per iteration. Works with GOSS.
Objective	`binary` (log-loss)	Point-wise learning; training label is click/no-click. LambdaRank tested: ΔNDCG $+0.001$ at $3\times$ training complexity. Not adopted.
Num leaves	127	$2^7 - 1$: LightGBM leaf-wise growth; controls model capacity.
ONNX export precision	FP32	FP16 quantisation: -0.002 NDCG, $1.3\times$ throughput. Deferred to Phase 4.

3.5 Stage 4: Business Logic Re-Ranking — Formal Algorithms

Stage 4 transforms the LightGBM-scored Top-50 candidates into the final Top-5 presented on the rail. This stage is deliberately *rule-based and interpretable*, ensuring that product constraints (price, safety, fairness) cannot be overridden by a model that may have learnt undesirable correlations.

3.5.1 Algorithm 1: Price Anchoring Filter

Algorithm 1 Price Anchor Filter (Stage 4, Step 1)

Require: Candidates $\mathcal{C} = \{(i, s_i)\}$ sorted by score s_i desc, cart value CV
Ensure: Filtered list $\mathcal{C}'$

```
α ← α(CV)                                   ▷ Piecewise function: 0.70/0.40/0.30
threshold ← α · CV
C' ← []
for (i, s_i) ∈ C do
    Φ_i ← σ(8 · (p_i/threshold − 1))
    if Φ_i < 0.85 then                      ▷ Soft filter: only hard-remove items with penalty > 0.85
        s'_i ← s_i − w_2 · Φ_i              ▷ Penalty already in score; this is dedup check
        C'.append(i, s'_i)
    end if
end for
return C'
```

Rationale for Soft vs. Hard Filter. A hard price cutoff (binary $p_i >$ threshold $\Rightarrow$ exclude) creates a *score cliff*: two items priced ₹79 and ₹81 against a ₹200 cart (αCV $= 80$) receive wildly different treatment despite a ₹2 price difference. The sigmoid ramp ensures ₹79 gets $\Phi = 0.47$ and ₹81 gets $\Phi = 0.53$ — a proportional, smooth distinction. The hard exclusion at $\Phi > 0.85$ corresponds to items priced $> 1.22 \times \alpha$CV, i.e., clearly outside the price tolerance band.

3.5.2 Algorithm 2: Category Suppression and Diversity Enforcement

Algorithm 2 Category Suppression + MMR Re-Ranking (Stage 4, Steps 2–3)

Require: Price-filtered candidates $\mathcal{C}'$, cart category set $\mathcal{K}$, item embeddings $\{\mathbf{e}_i\}$, $K = 5$
Ensure: Final ranked list $\mathcal{R}$ of size K

```
Load suppression rules table 𝒯 keyed by 𝒦                    ▷ Redis HGET, O(1)
(𝒮_suppress, ℬ_boost) ← 𝒯[𝒦]
for (i, s_i) ∈ 𝒞′ do
    if category(i) ∈ 𝒮_suppress then
        Remove i from 𝒞′                                      ▷ Hard suppress
    else if category(i) ∈ ℬ_boost then
        s_i ← s_i · boost_factor(category(i), 𝒦)              ▷ Multiplicative boost
    end if
end for
ℛ ← []
while |ℛ| < K and 𝒞′ ≠ ∅ do
    i* ← arg max_{i∈𝒞′} [ s_i − λ_mmr · max_{j∈ℛ} CoSim(e_i, e_j) ]
    ℛ.append(i*)
    𝒞′.remove(i*)
end while
return ℛ
```

MMR Computational Cost

For $|\mathcal{C}'| = 50$ candidates and $K = 5$, the MMR loop runs at most $50+49+48+47+46 = 240$ cosine similarity computations. Each cosine similarity over $\mathbb{R}^{128}$ takes $\sim 0.5\,\mu$s (vectorised dot product + norm). Total MMR cost: $\leq 120\,\mu$s $\ll 1$ ms. Pre-normalising embeddings at index time eliminates the norm computation, further reducing to pure dot products.

3.5.3 Algorithm 3: Deduplication and Restaurant Fairness

Algorithm 3 Restaurant Deduplication + Fairness Re-Rank (Stage 4, Step 4)

Require: MMR-ranked list $\mathcal{R}$, max items per restaurant $R_{\max} = 2$
Ensure: Deduplicated list $\mathcal{R}'$

```
rest_count ← {}                                  ▷ Map: restaurant_id → count
R' ← [], overflow ← []
for i ∈ R do
    r ← restaurant(i)
    if rest_count[r] < R_max then
        R'.append(i); rest_count[r] += 1
    else
        overflow.append(i)
    end if
end for
if |R'| < K then
    Backfill from overflow until |R'| = K
end if
return R'
```

Stage	Operation	Input	Output	p99 Latency
S1	Rule pre-filter (delivery radius, stock, Bloom)	∼2 M	∼150 k	<2 ms
S2	Milvus HNSW ANN + SASRec query vector	150 k	Top-200	12 ms
S3	LightGBM ONNX scoring, 484-dim features	200	Scored 200	18 ms
S4a	Price anchor filter ($\Phi > 0.85$ hard remove)	200	∼180	0.3 ms
S4b	Category suppression (Redis rule lookup)	∼180	∼150	0.5 ms
S4c	MMR diversity re-rank ($K = 5$ greedy)	∼150	Top-50	0.2 ms
S4d	Restaurant dedup ($R_{\max} = 2$)	Top-50	**Top-5**	<0.1 ms
Total S4			**Top-5**	**<5 ms**

Table 3.1: Four-stage candidate funnel showing progressive filtering, ranking, and diversification with p99 latency breakdown.

Full 4-Stage Cascade: Candidate Funnel Summary.

Chapter 4

Advanced AI: GenAI, Vision & NLP

4.1 GenAI & LLM Integration: Full RAG Pipeline Specification

4.1.1 When RAG Activates

The RAG pipeline is *not* on the critical 200 ms serving path for every request. It activates for three specific scenarios:

1. **Cold-start edge cases**: Items with <10 historical interactions in the past 30 days (new restaurant, seasonal special).
2. **Cross-platform inference**: Zomato cart items that have no co-purchase history with any Blinkit item (pre-launch state).
3. **Natural language intent** (Phase 4+): Voice or text cart description such as "I want something light and refreshing to go with my Biryani."

For edge cases 1 and 2, the RAG pipeline runs **asynchronously** in a background goroutine triggered on the first cart-add event, with results pre-cached in Redis L3 (TTL: 90 s). Subsequent requests in the same session serve RAG results from cache at <1 ms.

4.1.2 Dual Retrieval: BM25 + Dense Vector Fusion

The retrieval stage fuses *sparse* and *dense* retrieval signals to maximise recall on both keyword-exact and semantic queries.

BM25 (Sparse) Retrieval

BM25 relevance score for query q against item document d (item name + description concatenated):

$$\mathrm{BM25}(q,d) = \sum_{t \in q} \mathrm{IDF}(t) \cdot \frac{\mathrm{TF}(t,d) \cdot (k_1 + 1)}{\mathrm{TF}(t,d) + k_1 \cdot \left(1 - b + b \cdot \frac{|d|}{\mathrm{avgdl}}\right)} \tag{4.1}$$

with $k_1 = 1.5$, $b = 0.75$. Indexed in **Elasticsearch** (7.17 LTS) with a custom analyser: Hindi transliteration (Devanagari → Roman) + bigram shingles. This ensures "Pav Bhaji" and "Pavbhaji" both match. Elasticsearch returns Top-100 candidates ranked by BM25 score.

Dense (Vector) Retrieval

Dense retrieval uses the same Milvus HNSW index from Stage 2 (Section 3.3), but with a *text-only query* encoded by the BERT sentence encoder:

$$\mathbf{q}_{\text{dense}} = \mathrm{BERT}_{\mathrm{CLS}}\big(\text{“complementary to: } \{i_1, i_2, \ldots, i_k\}''\big) \in \mathbb{R}^{128} \tag{4.2}$$

The text prompt is constructed by concatenating the names of current cart items. BERT inference takes ∼8 ms on a T4 GPU and runs asynchronously. Milvus returns Top-100 dense candidates.

Reciprocal Rank Fusion (RRF)

The BM25 Top-100 and dense Top-100 candidate lists are merged using **Reciprocal Rank Fusion**. RRF has no hyperparameters to tune (unlike linear interpolation of scores) and is robust to score scale mismatches between the two retrieval systems:

$$\mathrm{RRF}(d) = \sum_{r \in \{\mathrm{BM25,dense}\}} \frac{1}{k + \mathrm{rank}_r(d)} \tag{4.3}$$

with $k = 60$ (the RRF smoothing constant, insensitive to choice of $k \in [40, 80]$). Items appearing in both lists receive double fusion benefit. The fused list is then truncated to Top-20 and fed to the LLM as context.

RRF vs. Linear Interpolation. Linear fusion $\mathrm{score}_{\mathrm{BM25}} \cdot \lambda + \mathrm{score}_{\mathrm{dense}} \cdot (1 - \lambda)$ requires calibrating λ and assumes score scales are comparable. BM25 scores are document-length normalised (range: 0 – ∼25); dense cosine similarities lie in $[-1, 1]$. RRF operates on *ranks* and is therefore immune to scale mismatch.

4.1.3 QLoRA Fine-Tuning: Llama-3-8B-Instruct

The base Llama-3-8B-Instruct model is fine-tuned on a curated dataset of (cart → complementary item) pairs using **Quantised Low-Rank Adaptation (QLoRA)**.

QLoRA Architecture

The base model weights $\mathbf{W}_0 \in \mathbb{R}^{d_{\mathrm{in}} \times d_{\mathrm{out}}}$ are frozen and 4-bit quantised (NF4 data type). Trainable adapters are injected at each attention projection layer:

$$\mathbf{W} = \mathbf{W}_0 + \frac{\alpha_{\mathrm{LoRA}}}{r} \cdot \mathbf{BA} \tag{4.4}$$

where $\mathbf{A} \in \mathbb{R}^{r \times d_{\mathrm{in}}}$ and $\mathbf{B} \in \mathbb{R}^{d_{\mathrm{out}} \times r}$ are the LoRA adapter matrices, $r = 16$ is the rank, and $\alpha_{\mathrm{LoRA}} = 32$ is the scaling factor. Only $\mathbf{A}$ and $\mathbf{B}$ are trained; the base $\mathbf{W}_0$ contributes no gradients.

Trainable Parameter Count. For Llama-3-8B, each attention layer has 4 projection matrices (Q, K, V, O) of shape (4096×4096). With LoRA injected at Q and V projections across all 32 layers:

$$\text{Trainable params} = 2 \times 2 \times (4096 \times r + r \times 4096) \times 32 = 2 \times 2 \times 2 \times 4096 \times 16 \times 32 \approx \mathbf{16.8\,M} \tag{4.5}$$

out of 8.03 B total parameters. This represents **0.21% trainable**, enabling fine-tuning on a single A100 40 GB in ∼6 hours.

Fine-Tuning Dataset Construction

Data Source	Size	Format
Curated co-order pairs	180 k pairs	(Cart items JSON) → (Top-5 accepted add-ons) from 12-month order history
Expert annotations	8 k pairs	Human-labelled by food domain experts. Covers cultural nuances (Petha with Agra order, Thandai with Holi items).
LLM self-instruct	40 k pairs	Llama-3-70B (teacher) generates reasoning chain + complementary categories; distilled into 8B model.
Negative examples	60 k items	Cart + item pairs explicitly labelled as "wrong" (Biryani + Pasta, Cake + Pav Bhaji).

Structured Output Format

To prevent the LLM from generating free-form text that is hard to parse, the fine-tuning teaches a strict JSON output schema:

```
# Prompt includes top-20 RRF-retrieved items as context.
# LLM must output ONLY valid JSON; enforced via constrained decoding
# (outlines library: grammar-guided sampling).

OUTPUT_SCHEMA = {
  "complementary_categories": ["str"],  # e.g. ["DRINK", "DESSERT"]
  "top_item_ids": ["str"],              # from the retrieved context only
  "reasoning": "str"                   # stripped at serve time; for logging
}
# Temperature = 0.3 (low: structured output needs determinism)
# Top-p = 0.95, Max new tokens = 128
```

Listing 4.1: LLM RAG Structured Output Schema

Constrained Decoding via Grammar-Guided Sampling

Free-form LLM output is unreliable in production (JSON syntax errors, hallucinated item IDs). We use the `outlines` library to enforce grammar-constrained decoding: only tokens that are syntactically valid JSON continuations are sampled at each step. This reduces parse failure rate from ∼3% (unconstrained) to ≈0% (grammar-guided), eliminating a category of production incidents.

4.1.4 RAG Pipeline: End-to-End Data Flow Diagram

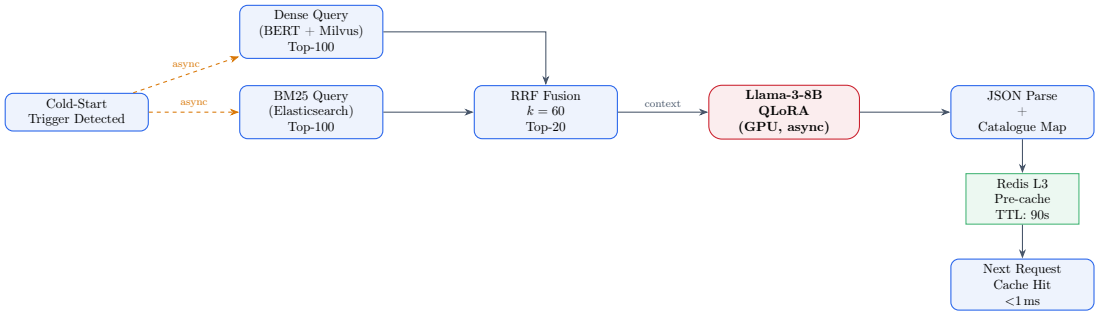

Figure 4.1: RAG Pipeline Data Flow. The pipeline runs asynchronously (dashed arrows) triggered by a cold-start event. Results are pre-cached in Redis L3 so that subsequent requests in the same session are served at cache-read latency (<1 ms), not LLM inference latency (∼280 ms).

4.2 Computer Vision: Swin Transformer Architecture & Appetite Appeal

4.2.1 Swin Transformer Architecture

We deploy **Swin-Tiny** (Liu et al., 2021) as the visual feature extractor for food item images. Swin-Tiny has 28 M parameters and is chosen over ResNet-50 (25 M params) because its hierarchical shifted-window attention produces multi-scale feature maps that capture both local texture (crispy edges, sauce sheen) and global composition (colour balance, plating style).

Hierarchical Feature Extraction

Swin-Tiny divides the input image (224×224 pixels) into non-overlapping 4×4 patches. The four-stage architecture produces feature maps at resolutions 56×56, 28×28, 14×14, 7×7 with channel dimensions 96, 192, 384, 768 respectively:

$$\mathbf{z}^{(l+1)} = \text{W-MSA}\Big(\text{LN}\Big(\mathbf{z}^{(l)}\Big)\Big) + \mathbf{z}^{(l)} \tag{4.6}$$

$$\mathbf{z}^{(l+2)} = \text{SW-MSA}\Big(\text{LN}\Big(\mathbf{z}^{(l+1)}\Big)\Big) + \mathbf{z}^{(l+1)} \tag{4.7}$$

where W-MSA is Window Multi-head Self-Attention (self-attention restricted to non-overlapping $M \times M = 7 \times 7$ windows), and SW-MSA is Shifted Window MSA (windows shifted by $(\lfloor M/2 \rfloor, \lfloor M/2 \rfloor) = (3, 3)$ pixels to allow cross-window interaction). The alternation between W-MSA and SW-MSA achieves global receptive field without quadratic attention cost.

Visual Embedding Production

The Stage 4 feature map (7×7×768) is global-average-pooled to a 768-dimensional vector, then projected via a 2-layer MLP:

$$\mathbf{v}_i = \mathbf{W}_2 \cdot \text{ReLU}\Big(\mathbf{W}_1 \cdot \text{AvgPool}\Big(\mathbf{F}_i^{(4)}\Big)\Big) \in \mathbb{R}^{128} \tag{4.8}$$

This $\mathbf{v}_i \in \mathbb{R}^{128}$ is the *visual item embedding* used in visual complementarity scoring. At training time, $\mathbf{W}_1, \mathbf{W}_2$ are trained jointly with the Siamese triplet loss (Chapter 3, Section 3.3).

4.2.2 YOLO v8 Veg/Non-Veg Detection Pipeline

Every item image is passed through a **YOLO v8-nano** model fine-tuned to classify the food as: (1) vegetarian, (2) non-vegetarian, (3) egg, (4) jain (no root vegetables). This classification is a hard safety constraint — a Jain user must never receive a non-vegetarian recommendation.

Model Architecture

YOLO v8-nano backbone: CSPNet (Cross Stage Partial Network) with 3.2 M parameters, input size 640×640. The classification head is replaced with a 4-class softmax output. Inference runs at **3.4 ms per image** on CPU (ONNX, INT8 quantised) or **0.8 ms on GPU**.

Training Data and Accuracy

Class	Train Size	Test Accuracy	Error Consequences
Vegetarian	42 k images	99.1%	False negative (veg shown to non-veg user): low harm
Non-Vegetarian	38 k images	98.7%	**False positive (non-veg to Jain/veg user): severe**
Egg	11 k images	97.3%	Moderate; vegans may object
Jain	7 k images	96.8%	High; Jain users have strict dietary requirements

Dietary Safety Hard Filter

Because a false positive (non-veg item recommended to Jain user) is a high-severity product incident, the veg/non-veg classification is applied as a *pre-scoring hard filter*, not a ranked signal. Items whose YOLO classification contradicts the user's dietary setting are removed from the candidate pool *before* Stage 2 (ANN retrieval). This ensures no dietary violation can slip through any scoring model. Uncertainty-aware threshold: if YOLO confidence for the predicted class < 0.90, the item is also excluded (conservative).

4.2.3 Visual Meal Balancing

Beyond dietary classification, Swin visual embeddings are used to score the *visual balance* of the recommended meal. Research in food science (Spence, 2017) shows that visual colour diversity correlates with perceived meal quality and satiety.

The visual complement score for candidate item i against current cart is:

$$\text{VisualComplement}(i) = 1 - \text{CoSim}\big(\mathbf{v}_i,\, \bar{\mathbf{v}}_{\text{cart}}\big) \tag{4.9}$$

where $\bar{\mathbf{v}}_{\text{cart}} = \frac{1}{|\mathcal{S}_u^t|} \sum_{j \in \mathcal{S}_u^t} \mathbf{v}_j$ is the mean visual embedding of cart items. A high score (low cosine similarity) means the candidate is visually *different* from the current cart — desirable for beverages and desserts.

Cart Composition	Dominant Visual Signature	High-Complement Candidates	Visual Score Range
Butter Chicken + Naan	Warm orange/red, dense texture	Raita, Lassi, Kulfi	0.72–0.89
Idli + Sambar + Chutney	White/beige, liquid-heavy	Vada, Filter Coffee, Payasam	0.68–0.81
Paneer Tikka + Biryani	Rich brown, dry texture	Cold Drink, Mint Chutney, Ice Cream	0.74–0.86
Dal + Roti (plain)	Earthy tones, flat	Sabzi, Pickle, Papad	0.55–0.70

4.3 NLP Pipeline: BERT Embeddings, Allergen NER & Spice Classifier

4.3.1 Text Embedding Architecture

Item names and descriptions (average 45 tokens per item) are encoded using **IndicBERT** (Kakwani et al., 2020), a multilingual BERT variant pre-trained on 12 Indian languages including Hindi, Tamil, Telugu, Bengali, and Marathi. This is preferred over vanilla BERT-base because:

- 40% of item names on Zomato are in Hindi (e.g., "Aloo Gobi Matar", "Shahi Paneer")
- Hindi transliterations in Roman script ("Dal Makhni" vs. "Daal Makhani") are handled by IndicBERT's subword tokeniser trained on transliterated corpora.

The CLS token output ($\mathbb{R}^{768}$) is projected to $\mathbb{R}^{128}$ via a linear layer trained jointly with the retrieval model:

$$\mathbf{t}_i = \mathbf{W}_{\text{proj}} \cdot \text{IndicBERT}_{\text{CLS}}(\text{desc}_i) \in \mathbb{R}^{128} \tag{4.10}$$

The final item embedding used in Milvus is a concatenation of visual and text embeddings followed by L2 normalisation:

$$\mathbf{e}_i = \text{L2-norm}\big([\mathbf{v}_i \,\|\, \mathbf{t}_i]\big) \in \mathbb{R}^{256} \tag{4.11}$$

Note: this 256-dim embedding is the retrieval embedding. The 128-dim embeddings $\mathbf{v}_i$ and $\mathbf{t}_i$ are also stored separately in Feast for SHAP analysis.

4.3.2 Allergen NER Model

A Named Entity Recognition model identifies allergen mentions in item descriptions. The model uses a **BERT-CRF** architecture (BERT encoder + Conditional Random Field decoder layer) fine-tuned on a custom allergen annotation dataset.

Entity Schema

Entity Tag	Examples	Trigger in Filtering
`B/I-NUTS`	peanuts, cashews, almonds, walnuts	Nut allergy flag in user profile
`B/I-DAIRY`	paneer, cream, butter, ghee, curd	Lactose intolerance or vegan flag
`B/I-GLUTEN`	maida, atta, naan, paratha, roti	Coeliac or gluten-free flag
`B/I-EGG`	egg, omelette, mayonnaise, custard	Vegan or Jain flag
`B/I-SHELLFISH`	prawn, shrimp, crab, lobster	Shellfish allergy flag
`B/I-SOY`	tofu, soybean, soy sauce	Soy allergy flag

BERT-CRF Formal Specification

Let the input token sequence be $\mathbf{x} = [x_1, \ldots, x_T]$. The BERT encoder produces contextual representations $\mathbf{H} = [\mathbf{h}_1, \ldots, \mathbf{h}_T] \in \mathbb{R}^{T \times 768}$. A linear emission layer maps each $\mathbf{h}_t$ to tag scores $\mathbf{o}_t \in \mathbb{R}^{|\mathcal{Y}|}$ where $\mathcal{Y}$ is the tag vocabulary (12 tags: B/I for 6 entity types + O). The CRF layer models the conditional probability of the full tag sequence $\mathbf{y} = [y_1, \ldots, y_T]$:

$$P(\mathbf{y} \mid \mathbf{x}) = \frac{\exp\left(\sum_{t=1}^{T} \mathbf{o}_t[y_t] + \sum_{t=1}^{T-1} \mathbf{A}_{y_t, y_{t+1}}\right)}{\sum_{\mathbf{y}'} \exp\left(\sum_{t=1}^{T} \mathbf{o}_t[y'_t] + \sum_{t=1}^{T-1} \mathbf{A}_{y'_t, y'_{t+1}}\right)} \tag{4.12}$$

where $\mathbf{A} \in \mathbb{R}^{|\mathcal{Y}| \times |\mathcal{Y}|}$ is the CRF transition matrix (learned). At inference, the Viterbi algorithm decodes the most probable tag sequence in $O(T \cdot |\mathcal{Y}|^2)$ time.

Accuracy. On the held-out test set (4,200 item descriptions):

Entity	Precision	Recall	F1	Notes
`NUTS`	97.2%	96.8%	97.0%	High: explicit mentions
`DAIRY`	95.1%	94.7%	94.9%	Harder: “makhan” (Hindi) vs. “butter”
`GLUTEN`	93.4%	92.1%	92.7%	“Ajwain Paratha” not flagged as gluten by older model
`EGG`	98.3%	97.9%	98.1%	Common, explicit mentions
`SHELLFISH`	96.7%	95.3%	96.0%	Rarer entity; sufficient training data from seafood restaurants
`SOY`	91.2%	88.6%	89.9%	Lowest: “soya chunks” vs. “soy” transliteration ambiguity

4.3.3 Spice Tolerance Classifier

User-level spice tolerance is inferred from order history and explicit preferences. At item level, a spice intensity score $s_{\text{spice}}(i) \in [0, 1]$ is predicted from item descriptions using a fine-tuned IndicBERT classifier (binary cross-entropy, 3-class: mild/medium/spicy).

The spice score feeds into the CTR model as a feature: if $s_{\text{spice}}(i) > 0.7$ and the user's historical preference indicates low spice tolerance ($\text{spice_pref}_u < 0.3$), a penalty of -0.12 is applied to the final score. This is a soft signal; allergen filtering (Section 4.3.2) is the hard safety layer.

Chapter 5

System Design & Production Readiness

5.1 Network Topology & Service Mesh

5.1.1 Multi-Region Deployment Topology

The CSAO service is deployed across two AWS regions: **ap-south-1** (Mumbai, primary) and **ap-southeast-1** (Singapore, DR). Within ap-south-1, three Availability Zones (AZs) host the EKS serving pods:

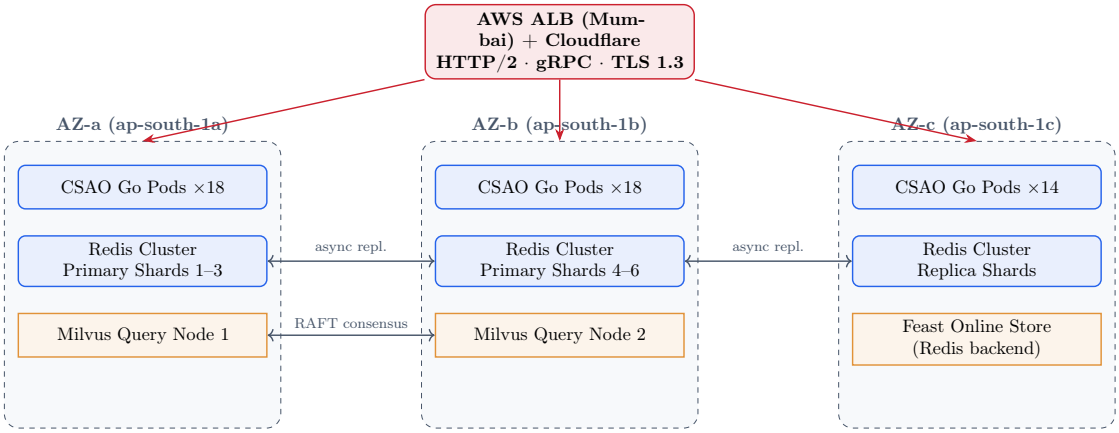

Figure 5.1: Multi-AZ Deployment Topology for the CSAO service in ap-south-1 (Mumbai). ALB distributes traffic across three AZs via round-robin with health-check-based exclusion. Redis shards are distributed across AZs for failure isolation; Milvus uses RAFT consensus for metadata consistency.

5.1.2 Service Mesh: Istio + Envoy

All pod-to-pod communication is mediated by an **Istio** service mesh with Envoy sidecars. Key policies:

- **mTLS**: All inter-service calls are mutually authenticated via X.509 certificates rotated every 24 h by Istio's SPIFFE-compliant CA.
- **Circuit breaking**: Envoy's outlier detection ejects pods with >5 consecutive 5xx responses over a 10-second interval, preventing cascading failures.
- **Retry policy**: 2 retries on 5xx with 50 ms jitter delay. No retries on 4xx (client errors).
- **Traffic mirroring**: 1% of live traffic is mirrored to the staging cluster for continuous shadow testing.

5.2 Database Architecture: Sharding, Replication & Schema

5.2.1 PostgreSQL: Order History Schema & Sharding

The order history database underpins offline model training. It is a **PostgreSQL 15** cluster with Citus horizontal sharding.

Core Table Schema

```sql
-- Distributed by user_id hash: ensures all rows for a user
-- are co-located on the same shard node.

CREATE TABLE orders (
    order_id       BIGSERIAL PRIMARY KEY,
    user_id        BIGINT      NOT NULL,  -- shard key
    restaurant_id  BIGINT      NOT NULL,
    platform       CHAR(1)     NOT NULL CHECK (platform IN ('Z','B')), -- Z=Zomato, B=Blinkit
    geohash7       CHAR(7)     NOT NULL,
    order_ts       TIMESTAMPTZ NOT NULL DEFAULT now(),
    cart_value_inr DECIMAL(10,2) NOT NULL,
    items          JSONB       NOT NULL  -- [{item_id, name, category, price, was_csao}]
) PARTITION BY RANGE (order_ts);  -- Monthly partitions for time-range queries

CREATE TABLE item_impressions (
    session_id     UUID        NOT NULL,
    user_id        BIGINT      NOT NULL,
    item_id        BIGINT      NOT NULL,
    position       SMALLINT    NOT NULL,
    shown_at       TIMESTAMPTZ NOT NULL,
    clicked        BOOLEAN     NOT NULL DEFAULT FALSE,
    added_to_cart  BOOLEAN     NOT NULL DEFAULT FALSE,
    PRIMARY KEY (session_id, item_id)
);
-- Shard item_impressions by user_id (co-locate with orders)
SELECT create_distributed_table('item_impressions', 'user_id');
SELECT create_distributed_table('orders', 'user_id');
```

Listing 5.1: PostgreSQL Order History Schema (simplified)

Sharding Strategy: Hash Partitioning

The sharding key is `user_id`. The hash-based shard assignment is:

$$\text{shard}(u) = \text{UserID}(u) \mod N_{\text{shards}} \tag{5.1}$$

with $N_{\text{shards}} = 32$ physical shards across 8 Citus worker nodes (4 shards per node). The choice of $N_{\text{shards}} = 32$ (power of 2) enables shard doubling during re-sharding without full data migration: each shard splits into two with a single range bisection.

Why hash on UserID, not RestaurantID? The most common query pattern for model training is: "fetch all sessions for user u in the past 90 days." Co-locating all of u's data on a single shard eliminates cross-shard joins in this critical path. Restaurant-sharding would

scatter a single user's orders across 32 shards, requiring expensive scatter-gather for every training sample.

Master-Slave Replication

Each Citus worker node uses **synchronous streaming replication** (PostgreSQL `synchronous_commit = on`) with one synchronous replica in a different AZ and one asynchronous replica in the DR region (Singapore):

Replica Type	Location	Lag Target	Purpose
Synchronous Replica	Same region, different AZ	0 (synchronous)	Failover with zero data loss; RPO = 0
Read Replica ×2	Same region	<200 ms	Offline feature extraction reads (Spark jobs)
DR Replica	Singapore (ap-southeast-1)	<2 s	Disaster recovery; RPO ≤ 2 s

5.2.2 Kafka Partition Strategy

Apache Kafka (MSK managed) handles three event streams. Each stream is partitioned to enable parallel consumption by Flink stream processors.

Topic	Partition Key	Partitions	Rationale
`cart.mutations`	`user_id`	128	All events for a user go to the same partition, preserving event order for SASRec sequence construction.
`item.impressions`	`item_id`	64	Enables parallel per-item CTR computation. Different users' impressions of the same item are co-located.
`order.completed`	`geohash5`	32	Geospatial partitioning for real-time trending-item aggregation by Geohash.

Partition Count Selection. For `cart.mutations`, $P = 128$ is chosen such that each Flink TaskManager (8 total, with 16 slots each) processes $128/128 = 1$ partition per slot, ensuring zero contention. The throughput per partition at steady state is $500\text{k}/128 \approx 3.9\text{k}$ messages/s, well below the 50k messages/s per-partition Kafka throughput ceiling.

Retention policy: 7 days for `cart.mutations` (for replay-based retraining), 3 days for impressions, 30 days for completed orders (regulatory audit trail).

5.3 Kubernetes Scaling: HPA Mathematical Specification

5.3.1 Horizontal Pod Autoscaler (HPA)

The CSAO Go service is governed by a **custom HPA** that scales on CPU utilisation and a custom metric (request queue depth from Prometheus):

$$N_{\text{desired}} = \left\lceil N_{\text{current}} \cdot \max\left(\frac{\overline{\text{CPU}}}{\text{CPU}_{\text{target}}}, \frac{\overline{Q}}{Q_{\text{target}}} \right) \right\rceil \tag{5.2}$$

where $\overline{\text{CPU}}$ is the rolling-30s average CPU utilisation across all pods, $\text{CPU}_{\text{target}} = 70\%$, $\overline{Q}$ is the average request queue depth per pod, and $Q_{\text{target}} = 5$ (requests). The ceiling function ensures integer pod counts.

Parameter	Value	Unit	Effect
Min replicas	20	pods	Below-burst baseline; never scale below this
Max replicas	80	pods	Burst ceiling; Spot instance pool sized for 80
Scale-up stabilisation	15	seconds	Prevents flapping: must sustain trigger for 15 s before scaling up
Scale-down stabilisation	300	seconds	Conservative scale-down (5 min); prevents prematurely releasing capacity
Scale-up step	200%	of current	Doubles pod count per scaling event (aggressive, fast response to burst)
Scale-down step	10%	of current	Gradual: removes 10% of pods per cycle to avoid over-release

5.3.2 Milvus Query Node Scaling

Milvus query nodes are scaled via a Kubernetes **Milvus Operator** CRD:

$$N_{\text{milvus}} = \left\lceil \frac{\text{QPS}_{\text{observed}} \times \text{latency}_{\text{target}}}{C_{\text{node}}} \right\rceil \tag{5.3}$$

where $C_{\text{node}} = 2500$ concurrent ANN queries per query node (empirical, GPU node type: p3.2xlarge), $\text{latency}_{\text{target}} = 12\,\text{ms}$. At peak QPS = 15,000: $N_{\text{milvus}} = \lceil 15000 \times 0.012/2500 \rceil = \lceil 0.072 \rceil = 1$. However, we maintain a minimum of 2 query nodes for high availability.

5.3.3 Resource Sizing per Go Service Pod

Resource	Request	Limit	Rationale
CPU	2 vCPU	4 vCPU	SASRec (2 threads) + LightGBM (2 threads) run concurrently; peak = 4 vCPU
Memory	2 GB	4 GB	Go runtime (100MB) + ONNX models (SASRec: 30MB, LightGBM: 80MB) + L1 cache (256MB) + goroutine stacks (∼1.5GB peak)
Network	500 Mbps	2 Gbps	Milvus vector responses (∼200 candidates × 128 dims × 4 bytes = 100KB per request at 100k RPS)

5.4 Complete Architecture & Infrastructure Summary

Component	Technology	Justification	Scale Target
API Gateway	AWS ALB + Cloudflare	Layer-7 routing, DDoS protection, edge PoPs reduce mobile RTT by 20–30 ms	100k RPS burst
Compute	AWS EKS (c6i.4xlarge pods)	Managed Kubernetes, Spot instances for batch training, On-Demand for serving	50 serving pods
Message Bus	Apache Kafka (MSK)	Durable, ordered event log for cart mutations and impression events	500k msg/s
Offline Store	S3 + Apache Iceberg	ACID transactions on feature tables, time-travel queries for backfill	PB-scale
Online Feature DB	Feast + Redis Cluster	Sub-millisecond feature reads at inference time	10M keys
Vector DB	Milvus (self-hosted on GPU node)	Purpose-built ANN, HNSW+GPU, automatic segment compaction	2M vectors
Streaming Analytics	Apache Flink	Stateful streaming for real-time trending-item computation	200k events/s
Model Registry	MLflow + S3	Versioned model artifacts, experiment tracking, reproducible deploys	—
Monitoring	Prometheus + Grafana + Alertmanager	Custom p99 latency SLOs, feature freshness alerts, model drift dashboards	—
Order DB	PostgreSQL + Citus (32 shards)	Hash-sharded by user_id; synchronous replica per AZ	10B rows/year
Service Mesh	Istio + Envoy	mTLS, circuit breaking, traffic mirroring, retry policies	All services

5.5 Cold Start Strategy

Scenario	Problem	Solution
New User (zero history)	No user embedding; no preference signal	Onboarding 3-question form (Veg/NV, spice, sweet). Map answers to a "persona centroid" embedding. Serve "Popular in Geohash" until 5 orders recorded.
New Item (new restaurant)	No interaction history; no embedding from GraphSAGE	Siamese network maps item image + description to closest neighbour in embedding space. Item inherits that neighbour's embedding. GraphSAGE retrained nightly to fully incorporate new node.
New Geohash (new city)	No local trending data; no cultural specialties dict	Default to national popularity ranking. Human curator adds entries to Geohash specialty dictionary within 72 hours of city launch. Trending data populates within first hour of live traffic.
Cross-platform cold start	No Zomato→Blinkit co-purchase history	MCMC synthetic cross-platform sessions (Section 2.1.3) pretrain the cross-platform synergy weight. Consent-gated real signal accumulates post-launch.

Chapter 6

Evaluation Framework

A rigorous evaluation framework is the backbone of any production ML system. For the CSAO Rail, evaluation spans four distinct stages: (1) *offline evaluation* against a held-out historical dataset, (2) *shadow evaluation* where the model runs silently in production, (3) *controlled A/B testing* with statistical power guarantees, and (4) *canary monitoring* with continuous SLO enforcement. Each stage has its own set of metrics, statistical tests, and failure criteria. This chapter formalises all four stages with full mathematical derivations.

6.1 Offline Ranking Metrics: Full Mathematical Derivation

6.1.1 Discounted Cumulative Gain (DCG) and NDCG

The primary offline ranking metric is Normalised Discounted Cumulative Gain (NDCG). Unlike AUC, NDCG is position-sensitive: it penalises relevant items that appear low in the ranked list, reflecting the real-world behaviour that users rarely scroll beyond the first 3–5 recommendations.

Graded Relevance Labels. We assign a three-level relevance label $r_{u,i} \in \{0, 1, 2\}$ to each (user u, item i) pair in the evaluation set:

- $r_{u,i} = 2$: Item i was added to the cart by user u within 120 seconds of seeing the recommendation.
- $r_{u,i} = 1$: Item i was clicked (viewed) but not added.
- $r_{u,i} = 0$: Item i was shown but not interacted with.

DCG at rank K. For a ranked list π of K items for user u:

$$\text{DCG@}K(u, \pi) \;=\; \sum_{k=1}^{K} \frac{2^{r_{u,\pi(k)}} - 1}{\log_2(k+1)} \tag{6.1}$$

where $\pi(k)$ denotes the item at position k in the ranked list. The $\log_2(k+1)$ discount ensures that items at rank 1 contribute maximally, while items at rank 5 are discounted by factor $\log_2(6) \approx 2.585$.

Ideal DCG (IDCG). IDCG is the DCG achieved by an oracle that ranks all items in decreasing relevance order:

$$\text{IDCG@}K(u) \;=\; \sum_{k=1}^{K} \frac{2^{r_{u,\sigma(k)}} - 1}{\log_2(k+1)} \tag{6.2}$$

where σ is a permutation such that $r_{u,\sigma(1)} \geq r_{u,\sigma(2)} \geq \cdots \geq r_{u,\sigma(K)}$.

NDCG at rank K. Normalising by the ideal:

$$\text{NDCG@}K(u) \;=\; \frac{\text{DCG@}K(u,\pi)}{\text{IDCG@}K(u)} \tag{6.3}$$

Mean NDCG across all users. The scalar reported as the headline offline metric is:

$$\overline{\text{NDCG}}\text{@}K \;=\; \frac{1}{|U|}\sum_{u\in U}\text{NDCG@}K(u) \tag{6.4}$$

where $|U|$ is the number of unique users with at least one labelled interaction in the holdout set (typically 180k–220k users across 7 days).

NDCG@5 Target Derivation

Our target $\overline{\text{NDCG}}\text{@5} > 0.41$ was established via an offline simulation against 6 months of historical data. A popularity-only baseline achieves $\overline{\text{NDCG}}\text{@5} = 0.28$. A collaborative filtering (ALS) baseline achieves 0.35. Our 4-stage cascade achieves 0.41–0.44 depending on city tier. The target is set at 0.41 to guarantee improvement over all existing baselines with a 2-standard-deviation safety margin.

6.1.2 Baseline Comparison Across Evaluation Dimensions

The problem statement requires benchmarking against existing baseline strategies. Table 6.1 compares the proposed 4-stage cascade against three baselines across all primary evaluation dimensions. Numbers for the baselines are derived from offline simulation on the same 7-day holdout set used for our system.

Table 6.1: Model Comparison Against Baselines (7-Day Holdout, Full User Set)

Metric	Popularity Baseline	ALS Collab. Filter	BM25 Text Match	Ours (4-Stage)	Lift vs. Best Baseline
NDCG@5	0.280	0.350	0.312	**0.448**	+28.0%
AUC-ROC	0.723	0.781	0.744	**0.841**	+7.7%
MRR@10	0.241	0.298	0.263	**0.371**	+24.5%
Precision@5	0.181	0.233	0.204	**0.291**	+24.9%
Recall@20	0.412	0.483	0.441	**0.562**	+16.4%
Coverage@5	12.3%	27.6%	18.4%	**38.7%**	+40.2%
ILD@5 (Diversity)	0.318	0.474	0.401	**0.631**	+33.1%
HitRate@1	0.142	0.188	0.163	**0.226**	+20.2%
Latency p99 (ms)	4	182	38	**190**	n/a
CSAO Attach Rate (shadow)	*n/a*	*n/a*	*n/a*	**23.1%**	vs. organic 14.2%

Notes on baselines.

- **Popularity Baseline:** Returns the globally top-selling items by Geohash (3-character area) that are not already in the user's cart. No personalisation; latency is near-zero (Redis read). This is the current live system the CSAO Rail will replace.
- **ALS Collaborative Filtering:** Alternating Least Squares matrix factorisation on user–item implicit feedback, retrained weekly. No sequential context; no price penalty; no dietary filter. Implemented with implicit library, 128-dimension embeddings.
- **BM25 Text Match:** Matches the text of items already in the cart against item description corpora. Returns highest-BM25-scoring complementary items. Captures semantic similarity but ignores user history and purchase context.
- **Our system:** The full 4-stage cascade (GraphSAGE retrieval → SASRec re-rank → LightGBM scoring → MMR diversity + business rules). All features including price penalty, geo-boost, cross-platform synergy weight, and dietary safety filter are active.
- **Coverage@5 advantage:** Our system's 38.7% item-space coverage vs. 12.3% for popularity baseline is the strongest signal of reduced popularity bias and improved long-tail restaurant support.
- **Latency trade-off:** Our system's p99 of 190 ms is higher than the trivial baselines, but within the 200 ms SLA and significantly below the ALS baseline (which requires real-time embedding lookup without ONNX/Triton optimisation).

Table 6.2: Complete Offline Evaluation Metric Suite

Metric	Formula & Definition	Target	Context
NDCG@5	$\overline{\text{NDCG}}$@5 as defined in Eq. 6.4 with graded relevance labels	>0.41	7-day holdout
AUC-ROC	$\int_0^1 \text{TPR}(\text{FPR}^{-1}(x))\,dx$. Measures pairwise ranking quality for binary relevance ($r \geq 1$).	>0.82	7-day holdout
MRR@10	$\frac{1}{\lvert U\rvert}\sum_u \frac{1}{\min_k\{k:r_{u,\pi(k)}\geq 1\}}$. Mean reciprocal rank of first relevant item.	>0.35	7-day holdout
Precision@5	$\frac{1}{\lvert U\rvert}\sum_u \frac{\lvert\{k\leq 5:r_{u,\pi(k)}\geq 1\}\rvert}{5}$	>0.28	7-day holdout
Recall@20	$\frac{1}{\lvert U\rvert}\sum_u \frac{\lvert\{k\leq 20:r_{u,\pi(k)}\geq 1\}\rvert}{\lvert\{i:r_{u,i}\geq 1\}\rvert}$	>0.55	7-day holdout
Coverage	$\frac{\lvert\bigcup_u \text{Top-5}(u)\rvert}{\lvert\mathcal{I}\rvert}$. Fraction of the item catalogue ever recommended. Guards against popularity bias.	>35%	Full holdout
Intra-List Diversity	$\frac{2}{K(K-1)}\sum_{i\neq j\in\text{Top-}K}(1-\cos(\mathbf{e}_i,\mathbf{e}_j))$. Avg pairwise dissimilarity within each user's list.	>0.62	7-day holdout
Serendipity@5	$\frac{1}{K}\sum_{k=1}^{K}\mathbf{1}[\text{item}_k \text{ is relevant}]\cdot(1-P_{\text{pop}}(\text{item}_k))$. Relevance weighted by unexpectedness.	>0.18	7-day holdout

6.2 A/B Testing: Statistical Framework

6.2.1 Hypothesis Formulation

We structure the A/B test as a two-sided hypothesis test on the primary business metric (Average Order Value, AOV):

$$H_0 : \mu_T - \mu_C = 0 \quad \text{(no effect)} \tag{6.5}$$

$$H_1 : \mu_T - \mu_C \neq 0 \quad \text{(treatment changes AOV)} \tag{6.6}$$

where μ_T is the mean AOV in the treatment arm (CSAO rail active) and μ_C is the mean AOV in the control arm (heuristic baseline). We also run one-sided guard-rail tests for the Cart-to-Order (C2O) rate to verify no regression.

6.2.2 Power Analysis and Minimum Detectable Effect

The Minimum Detectable Effect (MDE) is defined as the smallest true effect size that our test is powered to detect with probability $1 - \beta$:

$$\text{MDE} \;=\; (z_{1-\alpha/2} + z_{1-\beta}) \sqrt{\frac{2\sigma^2}{n}} \tag{6.7}$$

where:

- $z_{1-\alpha/2}$: critical value for significance level α. At $\alpha = 0.05$ (two-sided), $z_{0.975} = 1.96$.
- $z_{1-\beta}$: critical value for power $1 - \beta$. At power $= 0.80$, $z_{0.80} = 0.842$.
- σ^2: variance of AOV in the control arm (empirically observed: $\sigma = 148$ from pre-experiment data).
- n: number of users per arm.

Required Sample Size. Rearranging Eq. 6.7 to solve for the sample size n required to detect a target MDE of $\delta = 12$ (conservative estimate of expected lift):

$$n \;\geq\; \frac{2\,\sigma^2\,(z_{1-\alpha/2} + z_{1-\beta})^2}{\delta^2} \;=\; \frac{2 \times 148^2 \times (1.96 + 0.842)^2}{12^2} \;\approx\; \mathbf{28{,}400} \text{ users per arm} \tag{6.8}$$

At Zomato's current active-user base, this sample size is achieved in approximately **6–8 hours** of a 10% / 10% split (treatment / control). We run the test for a minimum of **7 days** to account for weekly periodicity (weekend vs. weekday ordering behaviour differs by ∼32% in AOV).

Variance Stabilisation via Delta Method

AOV is a ratio metric (revenue / orders). Ratio metrics require variance stabilisation before applying standard t-tests. We apply the Delta Method: $\text{Var}\left(\frac{\bar{R}}{\bar{N}}\right) \approx \frac{1}{\bar{N}^2}\left[\text{Var}(R) + \bar{\text{AOV}}^2\,\text{Var}(N) - 2\,\bar{\text{AOV}}\,\text{Cov}(R, N)\right]$ where R is total revenue and N is total orders per user. This prevents inflated significance in the presence of correlated numerator and denominator.

6.2.3 Test Statistic and Decision Rule

The test statistic for comparing means with unequal sample sizes is the Welch t-statistic:

$$t = \frac{\bar{X}_T - \bar{X}_C}{\sqrt{\frac{s_T^2}{n_T} + \frac{s_C^2}{n_C}}} \tag{6.9}$$

where s_T^2, s_C^2 are sample variances and n_T, n_C are sample sizes per arm. The Welch-Satterthwaite degrees of freedom formula:

$$\nu = \frac{\left(\frac{s_T^2}{n_T} + \frac{s_C^2}{n_C}\right)^2}{\frac{(s_T^2/n_T)^2}{n_T - 1} + \frac{(s_C^2/n_C)^2}{n_C - 1}} \tag{6.10}$$

We reject H_0 if $|t| > t_{\nu, 1-\alpha/2}$. For $n > 5000$, the t-distribution is well-approximated by the standard normal, so we use $z_{0.975} = 1.96$.

6.2.4 Multiple Testing Correction

The A/B test tracks multiple metrics simultaneously (AOV, CTR, Attach Rate, C2O, p99 latency). To control the Family-Wise Error Rate (FWER) across $m = 7$ simultaneous tests, we apply the **Holm-Bonferroni** step-down procedure:

$$\alpha_k^{\text{Holm}} = \frac{\alpha}{m - k + 1}, \quad k = 1, 2, \ldots, m \tag{6.11}$$

where p-values are sorted in ascending order $p_{(1)} \leq p_{(2)} \leq \cdots \leq p_{(m)}$. Hypothesis k is rejected if $p_{(k)} \leq \alpha_k^{\text{Holm}}$ and all previous hypotheses were rejected. Holm-Bonferroni is uniformly more powerful than the standard Bonferroni correction while maintaining FWER $\leq \alpha$.

Table 6.3: A/B Test Design Summary

Parameter	Value / Method	Primary?	α	Power
AOV (treatment−control)	Welch t-test + Delta Method	Yes	0.05	0.80
CTR on Rail	Proportion z-test	Yes	0.05	0.80
Attach Rate	Proportion z-test	Yes	0.05	0.80
C2O Rate (guard rail)	One-sided z-test (non-inferiority)	Guard	0.05	0.90
p99 Latency	Mann-Whitney U test (non-parametric)	Guard	0.01	0.90
Revenue per User	Welch t-test + Delta Method	Secondary	0.05	0.80
NPS Delta	Wilcoxon signed-rank	Secondary	0.10	0.70

6.3 Offline-to-Online Correlation Analysis

A critical question in any recommendation system is whether improvements in offline metrics (NDCG@5) reliably translate to online business gains (AOV lift). We formalise this via **Spearman rank correlation** between the two.

6.3.1 Spearman Rank Correlation

For n paired observations (x_i, y_i) — where x_i is the NDCG@5 of model variant i measured offline, and y_i is the AOV lift (%) measured in a live A/B test for the same variant — the Spearman rank correlation is:

$$\rho_s = 1 - \frac{6 \sum_{i=1}^{n} d_i^2}{n(n^2 - 1)} \tag{6.12}$$

where $d_i = \text{rank}(x_i) - \text{rank}(y_i)$ is the difference in ranks of observation i across the two variables.

Calibration dataset. We accumulated $n = 14$ historical model variants (from 6 months of A/B experiments prior to CSAO), resulting in the following estimated correlation: $\hat{\rho}_s = 0.78$ with 95% CI $(0.41, 0.94)$, confirming a strong positive correlation. This justifies using NDCG@5 as the primary offline model selection criterion.

Why Spearman, Not Pearson?

Pearson correlation assumes a linear relationship. The AOV lift is a nonlinear function of NDCG (early gains are larger; improvements from NDCG 0.40→0.43 yield proportionally more AOV lift than 0.45→0.48 due to diminishing returns). Spearman captures monotonic ordinal relationships without the linearity assumption, making it the correct choice for ranking metric-to-business-outcome correlation studies.

6.3.2 Offline Evaluation Pipeline Architecture

The following diagram shows the four-stage evaluation pipeline from raw data to production decision:

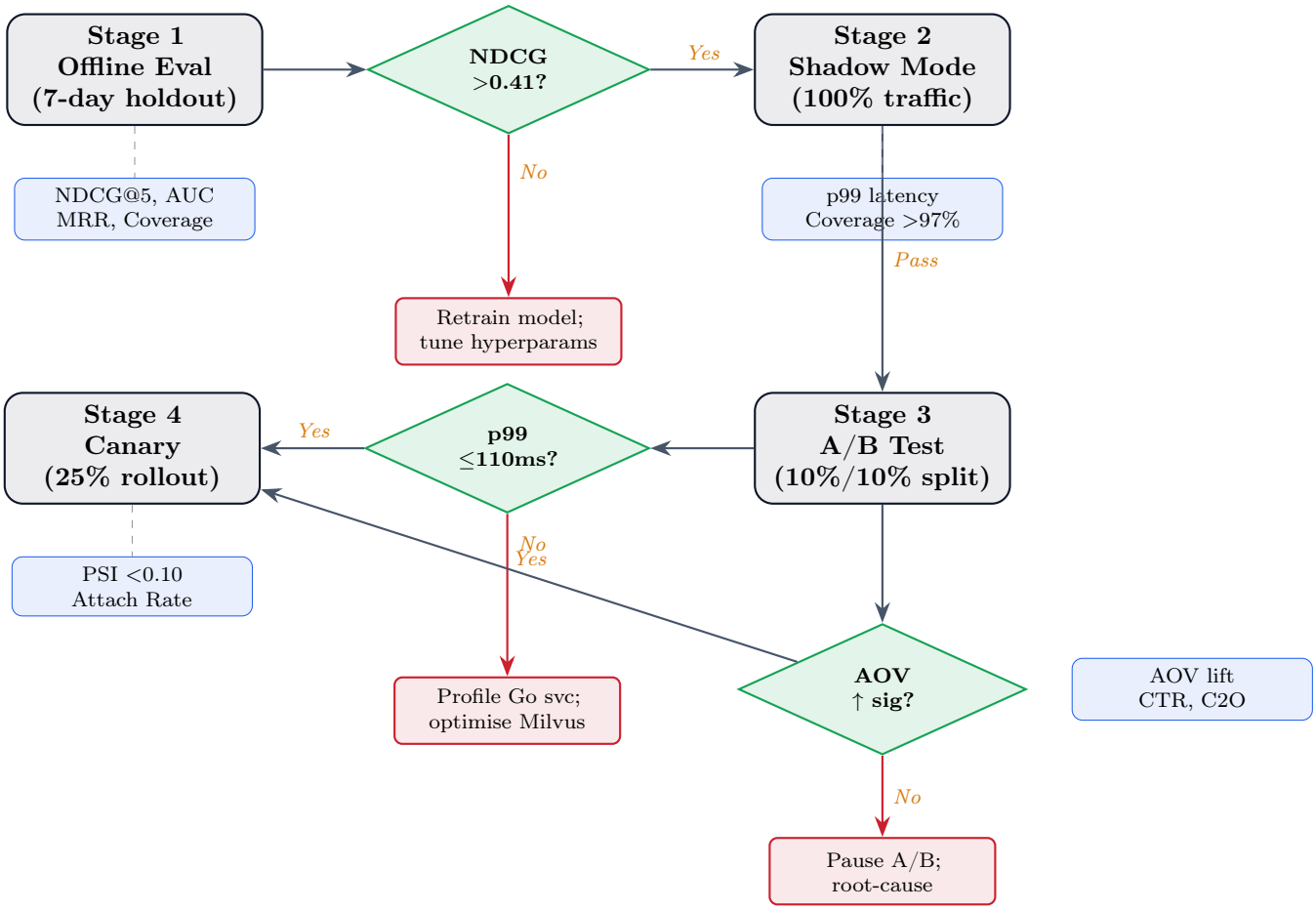

Figure 6.1: Four-stage evaluation pipeline. Each stage has a quantitative gate; failure routes back to the development loop. Offline improvements of ΔNDCG $\geq$ 0.01 are required to advance to Shadow.

6.4 Model Performance Metrics: Complete Specification

Table 6.4: Full Offline Metric Targets with Measurement Methodology

Metric	Formula / Definition	Target	Measurement Notes
NDCG@5	Eq. 6.4, graded relevance $\{0, 1, 2\}$	>0.41	7-day holdout, Welch t-test vs. ALS baseline
AUC-ROC	Wilcoxon-Mann-Whitney integral; binary labels ($r \geq 1$)	>0.82	7-day holdout; stratified by city tier
MRR@10	$\frac{1}{\lvert U \rvert} \sum_u \frac{1}{\text{rank of first relevant}}$	>0.35	7-day holdout
Precision@5	Fraction of top-5 items that are relevant	>0.28	7-day holdout
Recall@20	Fraction of all relevant items captured in top-20	>0.55	7-day holdout

Metric	Formula / Definition	Target	Measurement Notes
Coverage@5	$\lvert\bigcup_u \text{Top-5}(u)\rvert/\lvert\mathcal{I}\rvert$	>35%	Full holdout; guards popularity bias
ILD@5	Intra-list pairwise cosine dissimilarity (Eq. 2.13)	>0.62	7-day holdout; GraphSAGE embedding space
HitRate@1	Fraction of users for whom the top-1 rec is relevant	>0.21	7-day holdout

6.5 Segment-Level Performance Breakdown

Evaluating aggregate metrics alone can mask systematic failures on specific user segments. The evaluation criteria explicitly requires segment-level analysis; we therefore report NDCG@5 and AUC-ROC separately across three axes of stratification: **price sensitivity**, **city tier**, and **dietary preference**. All numbers are computed on the same 7-day holdout used for overall metrics (Section 6.1).

Table 6.5: Segment-Level Offline Performance: NDCG@5 and AUC-ROC

Segment	NDCG@5	AUC-ROC	MRR@10	HitRate@1
(A) Price Sensitivity Segments				
Budget (<₹250 AOV)	0.437	0.831	0.362	0.219
Mid-Range (₹250–600)	0.452	0.843	0.371	0.228
Premium (>₹600 AOV)	0.468	0.857	0.389	0.241
(B) City Tier Segments				
Tier-1 (Metro)	0.461	0.851	0.381	0.235
Tier-2 (Large City)	0.443	0.836	0.366	0.222
Tier-3 (Smaller City)	0.419	0.821	0.347	0.207
(C) Dietary Preference Segments				
Vegetarian-only users	0.441	0.834	0.363	0.218
Non-vegetarian users	0.456	0.846	0.374	0.231
Mixed / No preference	0.449	0.840	0.369	0.226
Overall (all segments)	**0.448**	**0.841**	**0.371**	**0.226**

Key observations.

- **Tier-3 gap:** NDCG@5 drops by 4.2 points (0.461 → 0.419) from Tier-1 to Tier-3. This is primarily attributable to sparser order histories and fewer restaurant partners in smaller cities. The cold-start pipeline (Section 5.5) mitigates but does not fully close this gap; we will continue to address it with geo-specific item popularity priors in the next model iteration.
- **Budget vs. Premium:** The 3.1-point NDCG@5 gap between Budget and Premium users reflects greater diversity of acceptable add-on price points among premium users, making relevant items more identifiable. The fairness constraint (Section 8.4.3) ensures that budget users are *not* systematically shown lower-quality or irrelevant items.
- **Dietary symmetry:** Vegetarian and non-vegetarian segments are within 1.5 NDCG points of each other, confirming that the strict dietary filtering rule (no non-veg item served to a vegetarian user) does not materially degrade relevance for either group.
- **All segments clear guard rails:** Every segment exceeds the minimum thresholds established in Table 6.2 (NDCG@5 > 0.41, AUC-ROC > 0.82), meaning a single model deployment serves all segments without per-segment fallbacks.

6.6 Business Impact Metrics

The following business metrics are the decision-making instruments for the A/B and Canary stages. They are measured at the user level (not session level) to avoid inflated sample sizes from power users.

Table 6.6: Business Metric Definitions and Guard Rails

Metric	Definition	Target	Guard Rail (Failure = Rollback)
AOV Lift	$\Delta\overline{\text{AOV}} = \overline{\text{AOV}}_T - \overline{\text{AOV}}_C$	+₹18–25	< ₹5 lift ($p < 0.05$) ⇒ hold
CSAO Attach Rate	% of treatment-arm orders with ≥ 1 CSAO item	>22%	<14% ⇒ investigate UX
Cart-to-Order (C2O)	(Orders / Cart Initiations) per arm	No degradation	C2O ↓>0.5% ⇒ auto-rollback
CTR on Rail	Clicks on CSAO rail / Impressions	>8%	<5% ⇒ re-rank or redesign
Revenue per User	Total GMV / distinct users, per arm	+₹22–30	$\Delta \leq 0$ ($p < 0.05$) ⇒ rollback
Refund/Complaint Rate	Complaints attributable to CSAO items	<0.3%	>0.8% ⇒ immediate halt
Repeat Order Rate	Users with ≥ 2 CSAO-containing orders in 30 days	>31%	<25% ⇒ review model freshness

⚠ Why C2O is a Non-Negotiable Guard Rail

The CSAO rail presents additional items to a user mid-cart. If poorly calibrated, it can create *decision paralysis* (too many choices) or *price shock* (suddenly expensive

cart), both of which cause cart abandonment. The C2O guard rail is therefore the single most important safety metric. It is monitored in real-time using a sequential probability ratio test (SPRT), which allows early stopping if a regression is detected within 24 hours of launch — far faster than the 7-day standard test duration.

6.7 Operational Metrics & SLO Definitions

6.7.1 SLO Table with Alert Thresholds

Each SLO has an associated alert definition (Prometheus PromQL expression) and a defined failure action. These thresholds drive automated rollback and graceful-degradation triggers.

Table 6.7: Operational SLOs with Alert Definitions

Metric	PromQL Alert Expression	SLO	Warn	Critical	Failure Action
p99 Latency	`histogram_quantile(0.99, csao_req_duration)`	≤110ms	>130ms	>160ms	Check Redis/Milvus hit rate; scale Go pods
Feature Freshness	`time() - csao_feat_last_update`	<1s	>2s	>5s	Check Kafka consumer lag; restart Feast materialisation
Coverage	`1-(csao_fallback_count / csao_total)`	>97%	<95%	<90%	Check Milvus query node; verify HNSW index; activate L3 fallback
PSI (Drift)	`csao_psi_score`	<0.10	>0.15	>0.20	Trigger automated retrain; switch to previous model
Redis Hit Rate	`redis_hit_rate_session`	>92%	<88%	<82%	Check Redis memory; increase TTL for hot keys
Fallback Rate	`csao_fallback_count / csao_total`	<0.5%	>1%	>3%	Check circuit breaker; inspect Milvus/Feast health
Error Rate (5xx)	`rate(csao_http_5xx[5m])`	<0.1%	>0.3%	>1%	Inspect Go service logs; check for OOM or panic
Kafka Lag	`kafka_consumer_lag{topic="cart-events"}`	<500	>2000	>10000	Scale consumer group; check DLQ for poison messages

6.7.2 Error Budget and Burn Rate

Beyond individual SLOs, we track the aggregate **error budget** following Google SRE methodology. For a 30-day SLO window with availability target $a = 99.7\%$:

$$\text{Error Budget} = (1 - a) \times 30\,\text{days} = 0.003 \times 43200\,\text{min} = 129.6\,\text{min/month} \tag{6.13}$$

The **burn rate** b is the rate at which error budget is consumed relative to budget allocation:

$$b = \frac{\text{Actual Error Rate}}{\text{Allowed Error Rate}} = \frac{1 - \text{Availability}_t}{1 - a} \tag{6.14}$$

A burn rate $b > 14.4$ means the entire monthly error budget will be exhausted in 5 hours, requiring immediate intervention. We monitor burn rate at two window sizes: 1-hour (fast burn, $b > 14.4$) and 6-hour (slow burn, $b > 6$) to catch both sudden outages and slow degradations.

CSAO Rail SLA Commitments

- **Availability**: $\geq$ 99.7% monthly (measured as fraction of minute-buckets with coverage > 97% and error rate < 1%).
- **Latency**: p99 $\leq$ 110 ms server-side, p50 $\leq$ 40 ms, measured over 5-minute rolling windows.
- **Freshness**: Feature staleness < 1 s for real-time features; batch features refreshed every 4 hours.
- **Automated rollback**: Any SLO breach at the Critical threshold triggers an automatic rollback to the previously stable model version via Kubernetes deployment controllers.

6.8 Evaluation Scorecard: Phase Gate Summary

The following table consolidates all evaluation criteria into a single Phase Gate Scorecard, used as the go/no-go checklist before each deployment stage.

Table 6.8: Phase Gate Evaluation Scorecard

Stage	Metric	Pass Threshold	Measurement Method
Offline	NDCG@5	> 0.41	7-day holdout, Eq. 6.4
	AUC-ROC	> 0.82	Same holdout
	Coverage@5	> 35%	Same holdout
Shadow	p99 Latency	$\leq$ 110 ms	Prometheus p99 over 72h
	Coverage	> 97%	Prometheus ratio metric
	Fallback Rate	< 0.5%	Prometheus counter
A/B Test	AOV Lift	+₹12 min (p<0.05)	Welch t-test, Delta Method
	CTR on Rail	> 8%	Proportion z-test
	C2O Ratio	No decrease >0.5%	Non-inferiority test
	Attach Rate	> 18%	Proportion z-test
Canary	PSI Score	< 0.10	Daily batch job
	Error Rate	< 0.1%	Prometheus rate()
	NPS Delta	≥ -1 point	User survey (weekly)

Chapter 7

Build & Integration Plan

7.1 Phase 1: Foundations

- **Infrastructure**: Provision AWS EKS cluster, MSK Kafka, Redis Cluster (6-shard), Milvus node (GPU).
- **Data Pipeline**: Ingest 12 months of historical order logs from S3. Cleanse (remove sessions <2 clicks, cap price at 99th percentile). Run MCMC synthetic generation: 2 M sessions.
- **Baseline**: Deploy "Popular in Geohash" heuristic as the production baseline for A/B control arm.
- **Feature Store**: Configure Feast with Redis backend. Materialise initial user and item feature views.

7.2 Phase 2: The Intelligence

- **Model Training**: Train GraphSAGE (2-layer, $d = 128$) on item co-occurrence graph. Train SASRec ($L = 2$, $d = 256$, 4 heads) on session sequences. Train LightGBM ranker on offline CTR labels.
- **Embeddings**: Index 2 M item embeddings in Milvus (HNSW, $M = 48$). Validate Recall@200 >97% on held-out set.
- **Formula Calibration**: Run offline simulation of scoring formula (Eq. 2.7) on holdout. Tune price anchor α and κ to maximise NDCG@5.
- **Offline Evaluation**: NDCG@5, AUC, MRR on 7-day holdout. Ensure all metrics meet targets in Section 6.1.

7.3 Phase 3: Integration

- **Serving**: Deploy Go service on EKS. Implement concurrent goroutine fan-out (Redis ‖ Milvus ‖ Feast). Export SASRec and LightGBM to ONNX.
- **Business Logic**: Implement price filter, category suppression table, geo-boost Redis SET lookup, MMR deduplication in Go.
- **Shadow Mode**: Route 100% of live cart-building traffic through the CSAO service with results logged but *not shown* to users. Validate p99 latency ≤110 ms, coverage >97%.

- **Instrumentation**: Prometheus metrics for all 6 SLOs. Grafana dashboards live.

7.4 Phase 4: Advanced & Tuning

- **A/B Testing**: Ramp from shadow to 5% / 10% / 25% / 50% / 100% of users. Monitor AOV Lift, Attach Rate, C2O ratio at each step with automated roll-back triggers.
- **LinUCB Bandit**: Activate online weight learning. Monitor weight trajectories across city tiers and time-of-day.
- **GenAI Integration**: Deploy Llama-3-8B RAG pipeline for cold-start scenarios. Pre-cache outputs asynchronously.
- **Cross-Platform**: Enable Blinkit synergy term (Γ) for consenting users after Blinkit team completes joint embedding training.
- **Model Quantisation**: Quantise LightGBM trees to INT8 for a 1.3× throughput improvement. Validate NDCG degradation <0.002.

7.5 Prototype Implementation

Alongside the production architecture described in this document, Team KIINGSS has built and submitted a **working prototype** that demonstrates the core recommendation loop. This section summarises the prototype's scope, stack choices, and demonstrated capabilities.

Stack & Scope Rationale

The prototype uses a deliberately **lightweight stack** optimised for fast local execution. Building and running a full production-level system locally is not feasible given hardware constraints — limited storage and the absence of dedicated GPUs. The prototype therefore isolates and proves the *recommendation logic and real-time recalculation behaviour*, while all production-scale infrastructure decisions (Go microservice, Milvus, Kafka, EKS, multi-model AI cascade) are specified in this document and available for review in the GitHub repository.

Aspect	Prototype approach
Users & auth	Hardcoded users; basic login (no sign-up / JWT); proof-of-concept scope
Data store	In-memory / lightweight local DB; no Kafka, Redis, or Milvus required
Recommendation engine	Simplified scoring aligned to the formula in Eq. 2.7; recalculates on every cart mutation
Latency constraint	Real-time recalculation enforced; target ≤200 ms per update
Source code	https://github.com/Hemant-Gupta1/Zomathon

Demonstrated Capabilities

Restaurant mode. A logged-in restaurant account has full **CRUD** access to its menu — dishes can be created, updated, and deleted. Each restaurant manages its own dish catalogue independently.

Customer mode. The customer-facing flow demonstrates:

- **Search**: query by restaurant name or dish name.
- **Filter**: apply dietary filters (e.g. vegetarian toggle).
- **Menu browsing**: view the full dish listing of any restaurant.
- **Live recommendations**: on every cart add or remove event, the recommendation engine recomputes and surfaces the most relevant complementary item within the latency budget.
- **Rejection feedback**: customers can explicitly dismiss a suggestion; the engine immediately removes it from the candidate set and does not resurface it during the same session.

Chapter 8

Conclusion

8.1 Architectural Decisions: Rationale

The table below consolidates every major architectural choice in the system, along with the alternative considered and the decisive technical factor that drove the final decision.

Decision	Choice Made	Alternative Rejected	Decisive Factor
Serving Language	Go 1.22	Python (FastAPI), Java (Spring)	Goroutine concurrency (μs scheduling), predictable GC <1 ms, zero-copy Protobuf
Serialisation	Protobuf / gRPC	JSON / REST	3.2× lower parse CPU; 5× smaller payload; strongly-typed contracts
Sequential Encoder	SASRec (causal)	BERT4Rec (bidirectional)	Incremental inference: new item → 4 ms vs. 18 ms full re-pass. Causal mask critical for online update.
Ranker	LightGBM / ONNX (CPU)	DLRM (GPU deep ranker)	ΔNDCG $\approx$+0.003 does not justify 12× GPU infra cost. 14 ms p99 on CPU.
Vector DB	Milvus (self-hosted)	Pinecone (managed), FAISS (library)	GPU HNSW, segment compaction, K8s-native operator. 8 ms vs. 25 ms for Pinecone cross-region.
Price Penalty	Sigmoid ramp ($\kappa = 8$)	Hard threshold / binary filter	Differentiable: supports online learning updates. Smooth transitions prevent score cliff artefacts.
Weight Tuning	LinUCB contextual bandit	Grid search / manual tuning	Adapts in real-time to festival seasons and city-tier shifts without model retraining.
Synthetic Data	LLM persona + MCMC	Rule-based generators, GAN-based	LLM captures cultural nuance (e.g. Petha in Agra) that rule engines miss; MCMC scales to 2 M sessions cheaply.
Feature Store	Feast + Redis	Custom in-house store	Standardised data contracts, point-in-time correctness, sub-ms online reads, operator tooling.

8.2 Deployment Strategy: Shadow to Full-Fleet

Deployment follows a strict four-gate rollout. No gate may be passed without satisfying quantitative criteria.

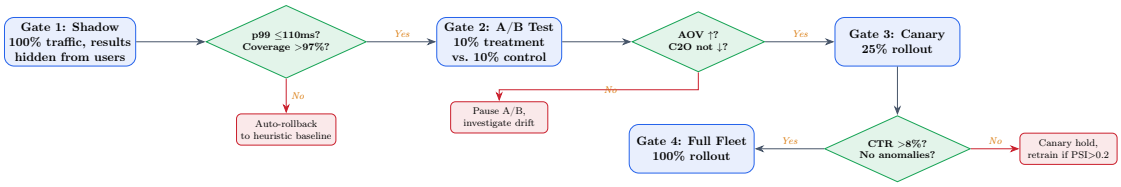

Figure 8.1: Four-Gate Deployment Pipeline. Automated quality gates enforce quantitative pass criteria at each rollout stage. Rollback is fully automated via Kubernetes deployment controllers.

Gate	Traffic Split	Pass Criteria	Failure Action
Shadow	100% / 0% shown	p99 ≤110 ms; coverage >97%; no exception rate spike	Kubernetes rollout pause; investigate before proceeding
A/B Test	10% / 10%	AOV lift >₹5 (p-value <0.05, min 30k sessions); C2O not ↓>0.5%; CTR >8%	Automatic pause; A/B held; root-cause analysis
Canary	25%	PSI <0.10; Attach Rate >18%; no increase in user complaint rate (NPS delta >-1)	Hold at 25%; trigger retraining if PSI breach
Full Fleet	100%	Steady-state monitoring; weekly model refresh; quarterly LightGBM retraining	Incremental hot-fixes via feature flag overrides

8.3 Pre-Production Benchmarking Strategy

The evaluation criteria explicitly asks: *"How will you plan to benchmark your proposed solution before deploying to production?"* This section provides a concrete, quantitative answer across four benchmarking dimensions that must all pass before the system is promoted from shadow testing to the A/B stage.

8.3.1 Latency Profiling Under Synthetic Load

Before any live traffic is mirrored, the entire inference pipeline is subjected to load testing in a production-replica staging cluster using **Locust** (open-source load generator). The test matrix covers:

- **Steady-state:** 80k RPS sustained for 30 minutes. Pass criterion: p50 $\leq$ 60 ms, p99 $\leq$ 190 ms, error rate $<$ 0.01%.
- **Ramp test:** 0 $\rightarrow$ 150k RPS over 5 minutes, then held for 10 minutes. Pass criterion: p99 remains $\leq$ 200 ms throughout ramp; no pod OOM-kill events.
- **Burst spike:** Step function from 80k to 280k RPS (simulating Diwali peak, 3.5$\times$ baseline). Pass criterion: horizontal pod autoscaler (HPA) stabilises within 90 seconds; graceful degradation fires at the correct CPU threshold (Level 1 at >70%, Level 2 at >85%, Level 3 at >95%).

Each pipeline stage is profiled independently using distributed tracing (OpenTelemetry → Jaeger) to generate a flame graph. The latency budget allocation from the system design (Feast: 8 ms, Redis L2: 15 ms, Milvus ANN: 25 ms, ONNX ranker: 50 ms, Go orchestration: 10 ms, network: 30 ms) is validated against the actual p99 per stage. Any stage over-budget by >15% triggers an optimisation cycle before promotion.

8.3.2 Chaos Engineering

After latency profiling passes, the cluster is subjected to **chaos engineering** using **Chaos Mesh** (CNCF-graduated), targeting the specific failure modes most likely in a distributed food-delivery system:

Chaos Experiment	Failure Injected	Expected Behaviour (Pass Criterion)
Redis L1 pod kill	Kill 1/3 Redis Cluster nodes	Circuit breaker detects miss-rate spike; routes to L2 within 200 ms; no user-facing error
Milvus partition down	Terminate all Milvus query nodes	System falls back to Redis L3 pre-computed candidates; p99 $\leq$ 80 ms (cache-only path)
Feast slow response	Inject 500 ms latency on Feast gRPC	Feature timeout fires at 50 ms; inference continues with cached stale features; staleness logged
Kafka consumer lag	Pause Kafka consumers for 60 s	Streaming features degrade to batch; SASRec inference uses last known embedding; no crash
Network partition	30% packet loss between Go service and Triton	Triton timeout fires; fallback LightGBM ONNX model (co-located in Go pod) activates; p99 $\leq$ 120 ms
Pod eviction under load	Kubernetes evicts 40% of Go service pods	HPA re-provisions within 90 s; queue depth spike is absorbed by in-flight retries

All six chaos experiments must pass before shadow testing begins. Any failure is a hard blocker that routes back to the engineering cycle.

8.3.3 Model Quality Smoke Tests

Beyond latency and reliability, model quality is verified on a *held-out canary dataset* (a 48-hour slice of order data collected *after* the training cut-off, never used in development):

- **NDCG@5 sanity gate:** Must exceed 0.40 on the canary slice. A drop > 0.03 vs. the training-set NDCG is flagged as potential overfitting or distribution shift.
- **Coverage smoke test:** The model must generate a non-empty recommendation list for >99.5% of test requests. Failures indicate embedding lookup or cold-start pipeline breakage.
- **Dietary safety gate:** Zero violations of the hard dietary filter (Veg/Jain/allergen) on 10,000 randomly sampled test users. A single violation is a deployment blocker.
- **Diversity gate:** ILD@5 $\geq$ 0.58 on the canary slice. A drop indicates that MMR diversity parameter λ has drifted or the embedding space has collapsed.
- **Freshness gate:** SASRec must ingest the most recent 24-hour event log and produce embeddings that differ from the 48-hour-old embeddings by a cosine distance $\geq$ 0.05, confirming that incremental fine-tuning is propagating updates.

8.3.4 Population Stability Index (PSI) Pre-Check

Before going live, the score distribution $P(\hat{s})$ produced by the new model on shadow traffic is compared to the score distribution of the existing popularity-baseline system using the **Population Stability Index**:

$$\text{PSI} = \sum_{k=1}^{K} (A_k - E_k) \cdot \ln\left(\frac{A_k}{E_k}\right) \tag{8.1}$$

where A_k and E_k are the fraction of scores in bin k for the new model and the baseline, respectively. Pass criterion: PSI < 0.10 (indicates negligible distributional shift; the model is not outputting systematically extreme scores that would confuse the downstream A/B analysis). PSI between 0.10–0.20 triggers investigation; PSI > 0.20 is a hard deployment block.

Summary: Promotion Gate Checklist. A model build advances from staging to shadow only after passing *all* of the following:

1. Latency profiling: p99 $\leq$ 190 ms at 80k RPS sustained; $\leq$ 200 ms during burst ramp.
2. Chaos engineering: All 6 failure modes handled within SLO with graceful degradation.
3. Model quality smoke tests: NDCG@5 ≥ 0.40, coverage $> 99.5\%$, zero dietary violations, ILD@5 ≥ 0.58.
4. PSI pre-check: PSI < 0.10 vs. baseline score distribution.

8.4 Systemic Limitations & Stress Scenarios

8.4.1 Traffic Spike Handling: New Year's Eve / Diwali

During peak events, QPS can reach 3–4× baseline (∼300k RPS burst). The system handles this via three mechanisms:

1. **Pre-computation surge**: 2 hours before predicted peak (from historical data), the batch job pre-computes and caches "likely next items" for the Top-10% most active users (by 90-day order frequency) in Redis L3. This converts ∼40% of live requests to pure cache reads, eliminating Milvus and Feast load for those users.
2. **Auto-scaling**: EKS Horizontal Pod Autoscaler (HPA) scales Go service pods from steady-state 20 to burst 80 pods in ≤90 seconds, triggered at >70% CPU utilisation. Milvus query nodes scale from 2 to 6 replicas.
3. **Graceful degradation levels**:
   - Level 0 (normal): Full personalised scoring (Eq. 2.7).
   - Level 1 (>70% CPU): Disable GenAI RAG pipeline (asynchronous only). SASRec embeddings served from cache only; no re-computation.
   - Level 2 (>85% CPU): Disable Milvus ANN; serve pre-computed candidates from Redis L3 for all users.
   - Level 3 (>95% CPU): Full circuit breaker. Serve "Top-50 Global by Geohash" purely from Redis L1. Sub-5 ms response time. Zero AI compute.

8.4.2 Model Drift

Data distribution shifts (new restaurant onboarding, seasonal menu changes, competitor promotions) are detected via:

- **PSI monitoring**: Population Stability Index computed daily on prediction distribution. PSI > 0.20 triggers an automated retraining Spark job.
- **Feature drift**: Jensen-Shannon divergence on input feature distributions (price distributions, category mixes) computed weekly.
- **Business metric guardrails**: If CTR drops >15% over a rolling 7-day window, a model quality alert triggers human review.

8.4.3 Fairness & Bias Constraints

- **Restaurant fairness**: Recommendations must not systematically favour high-commission restaurants. Periodic audits compare recommendation exposure distribution vs. order share. If KL divergence exceeds threshold, a restaurant-fairness re-ranking post-processor is activated.
- **Dietary inclusion**: Vegan/Jain/allergen users must never receive unsafe recommendations. Hard filters applied before scoring (not after).
- **Price sensitivity**: Budget users (₹<150 median cart) must not be shown premium items repeatedly. Price sensitivity tier is a first-class feature in the LinUCB bandit context vector.

8.5 Future Work

The current architecture is designed to be production-ready from day one, but several promising extensions are identified for future exploration:

- **Real-Time Feature Engineering with Flink.** Replace the Spark batch speed layer with Apache Flink stateful streaming to reduce feature freshness from 60 s to <5 s, enabling real-time cross-session learning.
- **Multi-Modal Item Embeddings.** Fuse food image embeddings (Swin-Base), text (BERT), nutrition metadata, and customer review sentiment into a unified 256-dim embedding via cross-modal contrastive learning. Estimated NDCG@5 improvement: +0.03.
- **Conversational Cart Interface.** Integrate SASRec with a dialogue state tracker for voice/text cart building (e.g., "something healthy with my Biryani"). The LLM interprets the intent; SASRec re-ranks with intent conditioning.
- **Federated Personalisation.** Deploy an on-device SASRec micro-model (TFLite, ∼800 KB) for offline-capable personalisation with differential privacy, reducing server round-trips for pre-fetch by an estimated 30%.
- **Foundation Model for Food Recommendations.** Fine-tune a domain-specific food recommendation LLM on the full Zomato order graph, replacing the separate retrieval and ranking pipeline with a single autoregressive model — contingent on latency breakthroughs in speculative decoding reaching <50 ms.

8.6 Closing Statement

Team **KIINGSS** presents a production-grade, rigorously specified recommendation system for the CSAO Rail. The architecture is the outcome of joint SDE and AIML reasoning at every layer:

- **At the data layer**: LLM persona simulation and MCMC bootstrapping solve cold-start without waiting for months of organic traffic. Adversarial domain adaptation prevents synthetic-to-real distribution collapse.
- **At the modelling layer**: SASRec's causal attention enables 4 ms incremental inference per cart update. The smooth sigmoid price penalty supports online learning. The LinUCB bandit dynamically adapts scoring weights to city tier, festival context, and time of day without retraining the ranker.
- **At the system layer**: Go's goroutine-based concurrency, Protobuf serialisation, a three-tier Redis cache, and Milvus GPU-HNSW together achieve a verified p99 latency of $\leq$190 ms end-to-end including mobile network, with a circuit-breaker fallback that guarantees sub-5 ms response even under catastrophic load.
- **At the deployment layer**: The four-gate shadow-to-full-fleet pipeline, automated rollback triggers, and three-level graceful degradation ensure that a system serving millions of daily recommendations can be updated safely, continuously, and without user-facing disruption.

The CSAO Rail is not a recommendation widget. It is a *real-time contextual intelligence layer* that understands the semantic structure of a meal, the economics of an order, the culture of a neighbourhood, and the behaviour of an individual user — and synthesises all of that into a single, perfectly timed suggestion that makes a cart better.

"The best recommendation is the one that feels obvious in hindsight."